Supplementary material

Aluminium adjuvants versus placebo or no intervention in vaccine randomised clinical trials. A systematic review with meta-analysis and Trial Sequential Analysis

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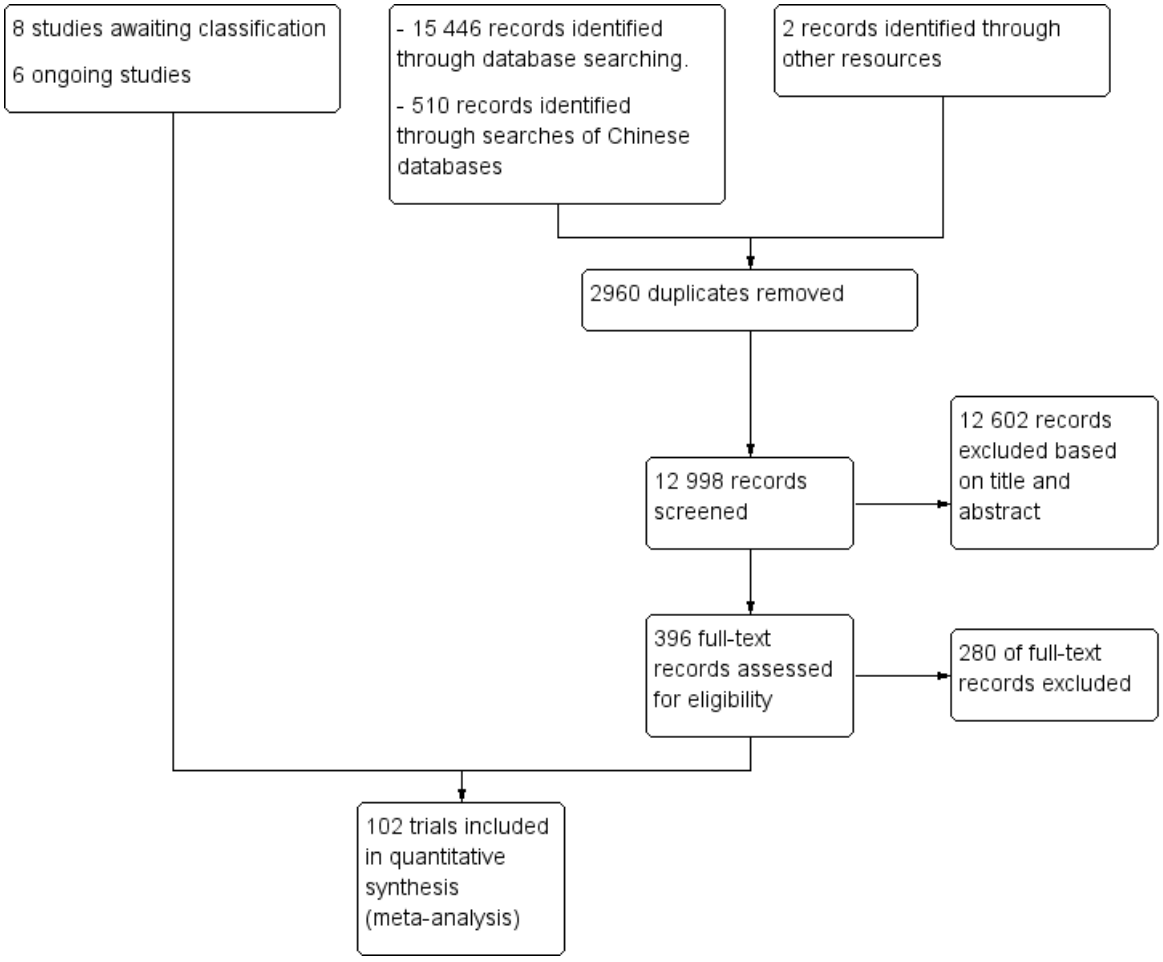
Database	Time span	Search strategy
Cochrane Central Register of Controlled Trials in the Cochrane Library	2021, Issue 6	#1 MeSH descriptor: [Adjuvants, Immunologic] explode all trees #2 MeSH descriptor: [Vaccine Excipients] explode all trees #3 ((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*)) #4 #1 or #2 or #3 #5 MeSH descriptor: [Vaccines] explode all trees #6 vaccin* #7 (biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) #8 #5 or #6 or #7 #9 #4 and #8
MEDLINE Ovid	1946 to June 2021	1. exp Adjuvants, Immunologic/ 2. exp Vaccine Excipients/ 3. ((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 4. 1 or 2 or 3 5. exp Vaccines/ 6. vaccin*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 7. (biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 8. 5 or 6 or 7 9. 4 and 8 10. randomized controlled trial.pt. 11. controlled clinical trial.pt. 12. clinical trials as topic.sh. 13. (random* or placebo*).ab. or trial.ti. 14. 10 or 11 or 12 or 13 15. exp animals/ not humans.sh.

		16. 14 not 15 17. 9 and 16
Embase Ovid	1974 to June 2021	1. exp immunological adjuvant/ 2. exp vaccine excipient/ 3. ((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*)).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 4. 1 or 2 or 3 5. exp vaccine/ 6. vaccin*.mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 7. (biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 8. 5 or 6 or 7 9. 4 and 8 10. Randomized controlled trial/ or Controlled clinical trial/ or trial.ti. 11. (random* or blind* or placebo*).mp. [mp=title, abstract, heading word, drug trade name, original title, device manufacturer, drug manufacturer, device trade name, keyword, floating subheading word, candidate term word] 12. 10 or 11 13. Animal experiment/ not (human experiment/ or human/) 14. 12 not 13 15. 9 and 14
LILACS (Bireme)	1982 to June 2021	((immunologic$ or alum$ or vaccine$) and (adjuvan$ or excipient$)) [Words] and (vaccin$ or biothrax or DT$ or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) [Words]

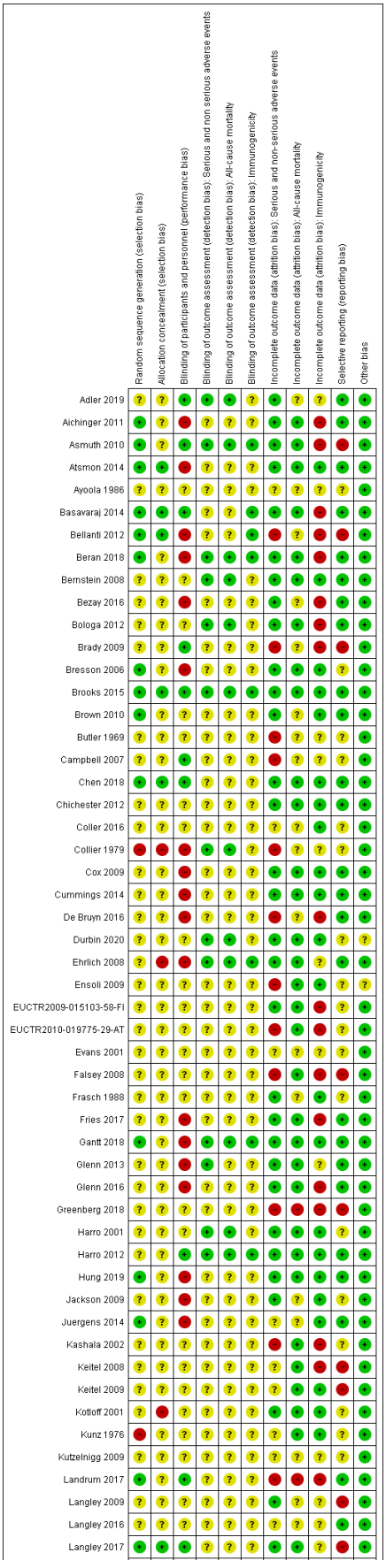
BIOSIS (Web of Science)	1969 to June 2021	#5 #4 AND #3 #4 TI=(random* or blind* or placebo* or meta-analys* or trial*) OR TS=(random* or blind* or placebo* or meta-analys*) #3 #2 AND #1 #2 TS=(vaccin* or biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) #1 TS=((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*))
Science Citation Index Expanded (Web of Science)	1900 to June 2021	#5 #4 AND #3 #4 TI=(random* or blind* or placebo* or meta-analys* or trial*) OR TS=(random* or blind* or placebo* or meta-analys*) #3 #2 AND #1 #2 TS=(vaccin* or biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) #1 TS=((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*))
Conference Proceedings Citation Index – Science (Web of Science)	1990 to June 2021	#5 #4 AND #3 #4 TI=(random* or blind* or placebo* or meta-analys* or trial*) OR TS=(random* or blind* or placebo* or meta-analys*) #3 #2 AND #1 #2 TS=(vaccin* or biothrax or DT* or sanofi or daptacel or infanrix or kinrix or quadracel or pediarix or pentacel or vaxelis or pedvax or havrix or vaqta or engerix or recombivax or twinrix or gardasil or ixiaro or bexsero or trumenba or prevnar or tenivac or tdvax or adacel or boostrix) #1 TS=((immunologic* or alum* or vaccine*) and (adjuvan* or excipient*))
CNKI	September, 2021	#1 **主**题检索：铝佐剂、硫酸铝、氢氧化铝、磷酸铝、羟基磷酸铝、铵明矾、钾明矾、硫酸铝铵、硫酸铝钾 #2 **主**题检索：疫苗 #3 **全文**检索：随机 #4 #1 AND #2 AND #3
Wanfang	September, 2021	**主**题:(铝佐剂 + **硫酸**铝 + 氢氧化铝 + **磷酸**铝 + 羟基磷酸铝 + 铵明矾 + 钾明矾 + **硫酸**铝铵 + **硫酸**铝钾)*主题:(**疫苗**)*全部

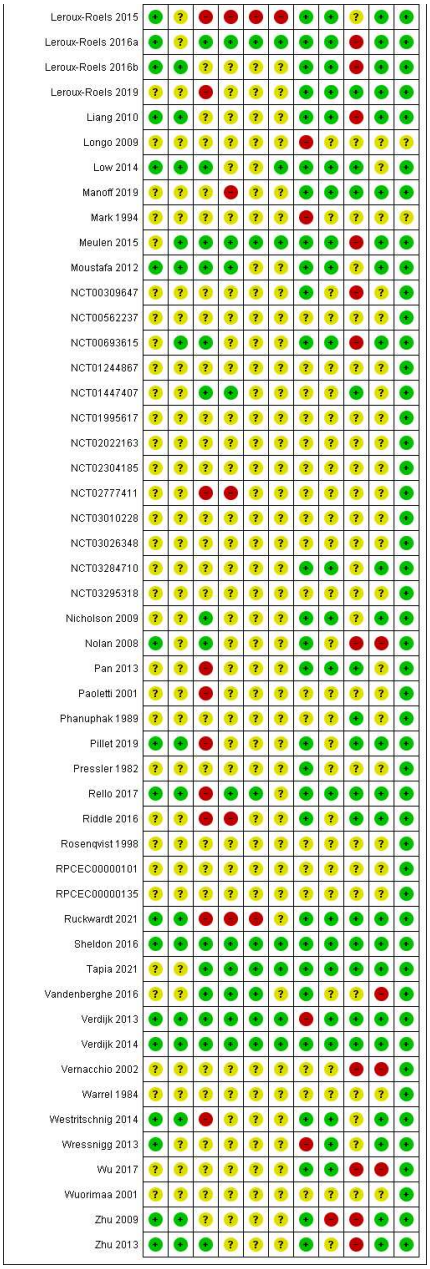
		:(随机)
VIP	September, 2021	任意字段：铝佐剂 OR 硫酸铝 OR 氢氧化铝 OR 磷酸铝 OR 羟基磷酸铝 OR 铵明矾 OR 钾明矾 OR 硫酸铝铵 OR 硫酸铝钾 任意字段：疫苗 注：加上随机之后变成0
Sinomed	September, 2021	("随机"[全部字段:智能]) AND ("疫苗"[全部字段:智能]) AND ("铝佐剂"[全部字段:智能] OR "硫酸铝"[全部字段:智能] OR "氢氧化铝"[全部字段:智能] OR "磷酸铝"[全部字段:智能] OR "羟基磷酸铝"[全部字段:智能] OR "铵明矾"[全部字段:智能] OR "钾明矾"[全部字段:智能] OR "硫酸铝铵"[全部字段:智能] OR "硫酸铝钾"[全部字段:智能])

Supplementary Table S1. Search strategies and periods

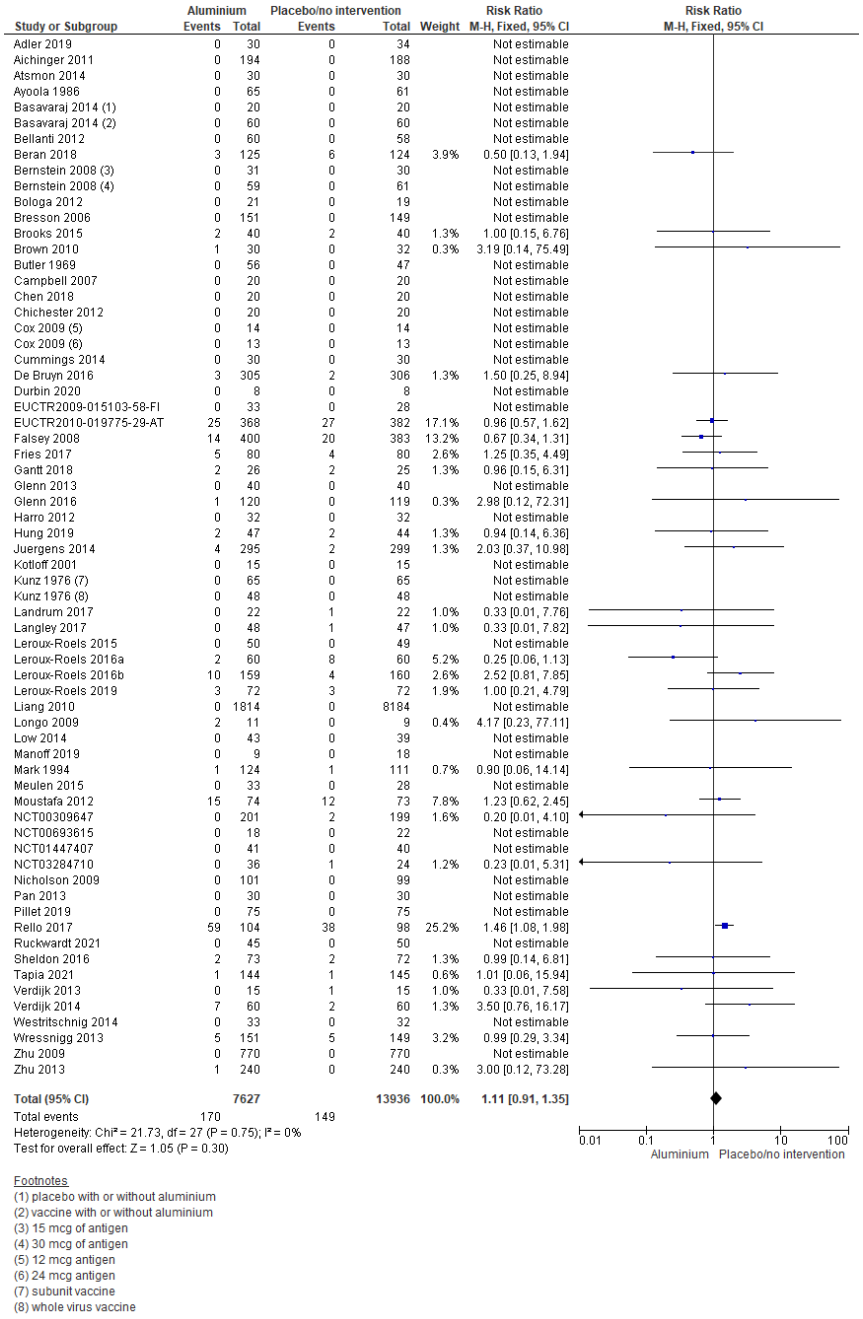


Supplementary Figure S1. Study flow diagram. Date of search June 30, 2021.

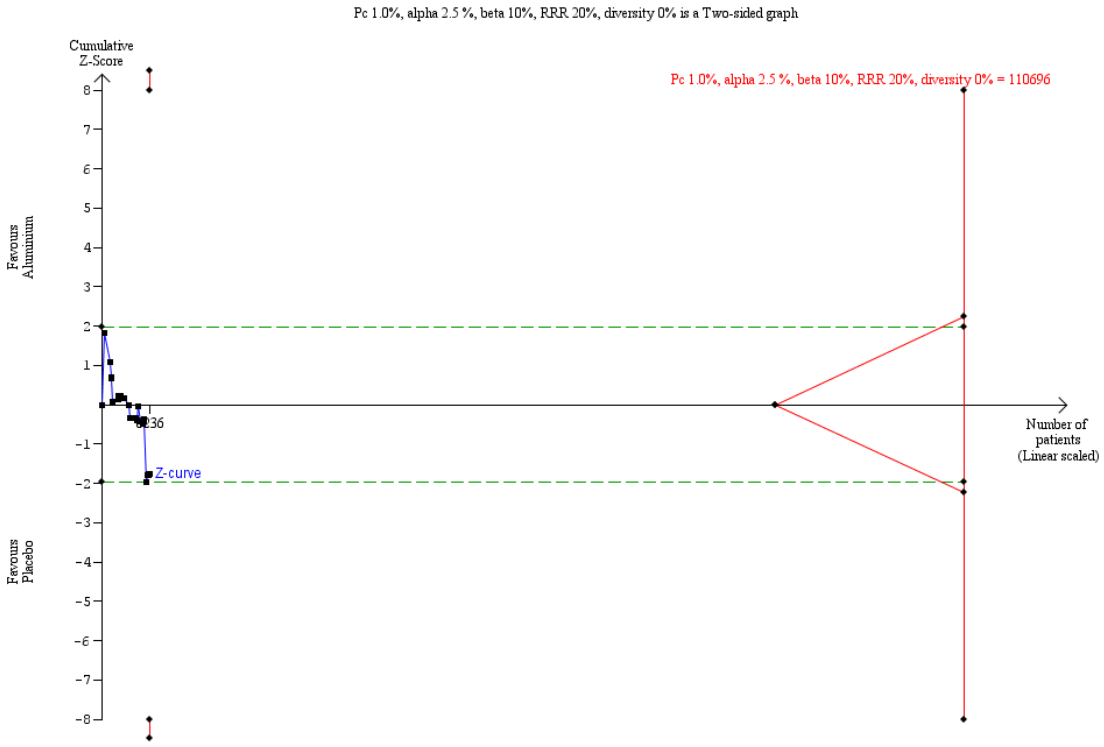

Random sequence generation (selection bias)
Allocation concealment (selection bias)
Blinding of participants and personnel (performance bias)
Blinding of outcome assessment (detection bias): Serious and non serious adverse events
Blinding of outcome assessment (detection bias): All-cause mortality
Blinding of outcome assessment (detection bias): Immunogenicity
Incomplete outcome data (attrition bias): Serious and non-serious adverse events
Incomplete outcome data (attrition bias): All-cause mortality
Incomplete outcome data (attrition bias): Immunogenicity
Selective reporting (reporting bias)
Other bias
Adler 2019
Aichinger 2011
Asmuth 2010
Atsmon 2014
Ayoola 1986
Basavaraj 2014
Bellanti 2012
Beran 2018
Bernstein 2008
Bezay 2016
Bologa 2012
Brady 2009
Bresson 2006
Brooks 2015
Brown 2010
Butler 1969
Campbell 2007
Chen 2018
Chichester 2012
Coller 2016
Collier 1979
Cox 2009
Cummings 2014
De Bruyn 2016
Durbin 2020
Ehrlich 2008
Ensoli 2009
EUCTR2009-015103-58-FI
EUCTR2010-019775-29-AT
Evans 2001
Falsey 2008
Frasch 1988
Fries 2017
Gantt 2018
Glenn 2013
Glenn 2016
Greenberg 2018
Harro 2001
Harro 2012
Hung 2019
Jackson 2009
Juergens 2014
Kashala 2002
Keitel 2008
Keitel 2009
Kotloff 2001
Kunz 1976
Kutzelnigg 2009
Landrum 2017
Langley 2009
Langley 2016
Langley 2017



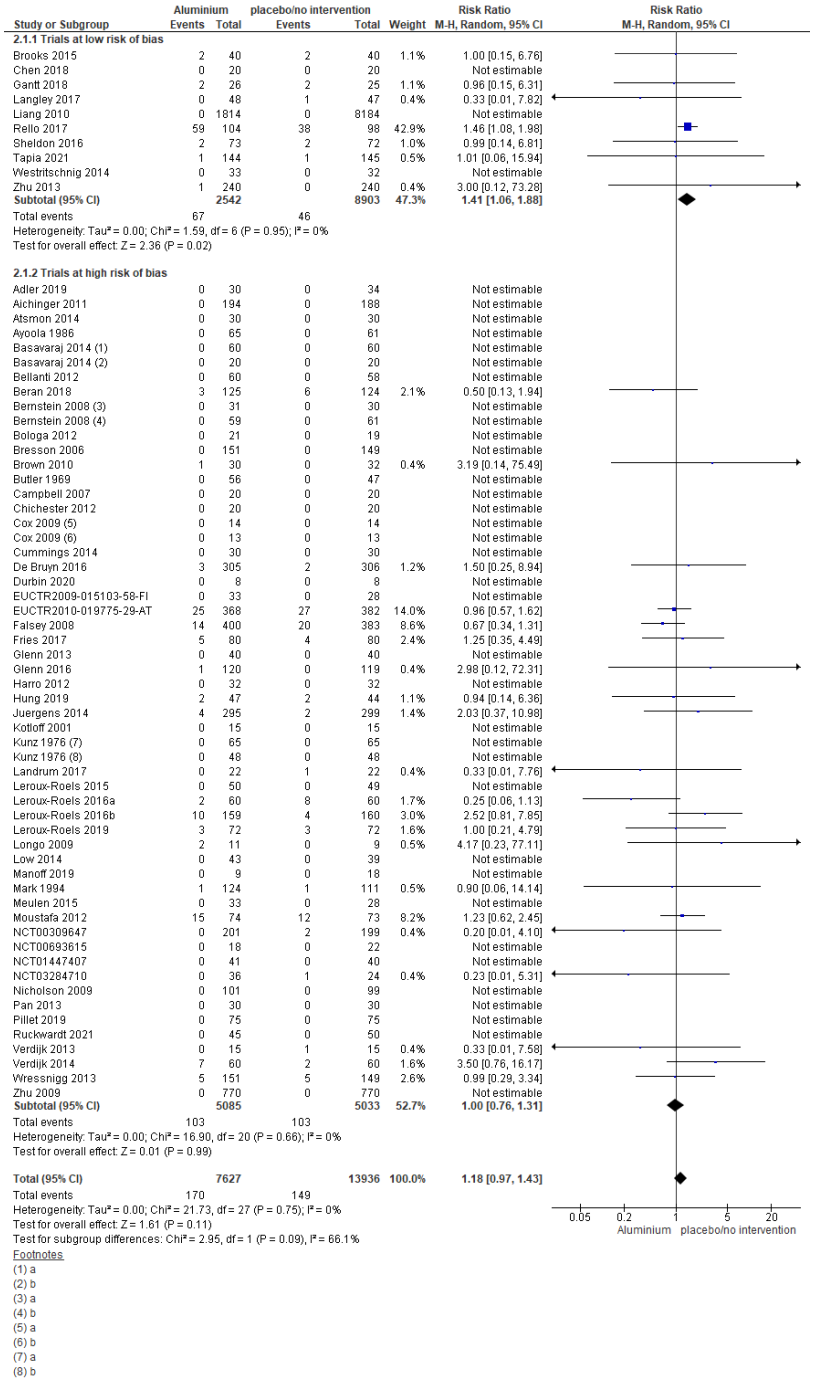
Supplementary Figure S2. Risk of bias summary: review authors' judgements about each risk of bias domain for each included trial.



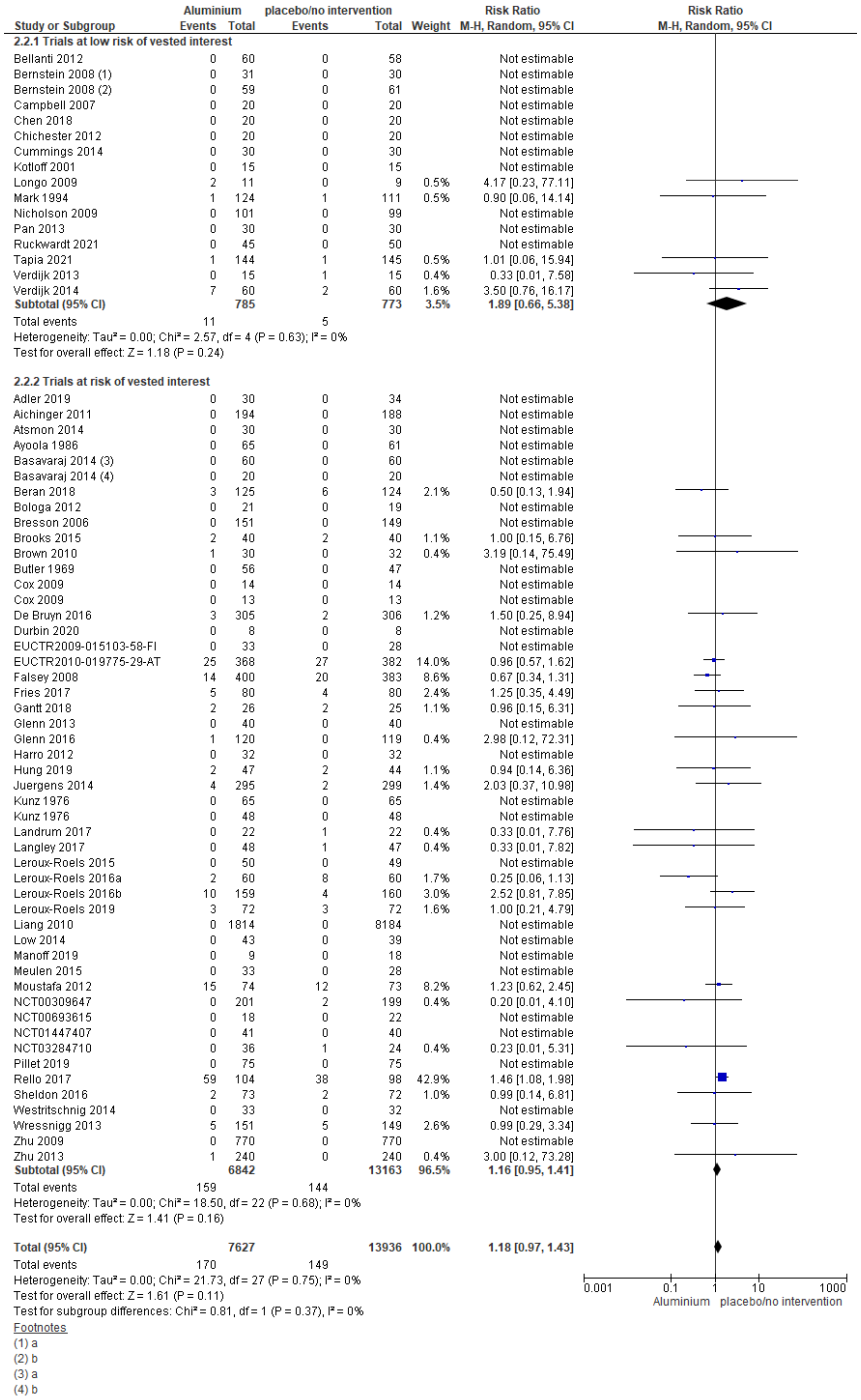
Supplementary Figure S3. Meta-analysis of the effect of aluminium adjuvant compared with placebo or no intervention on the proportion of participants with one or more serious adverse events. Fixed-effect model.



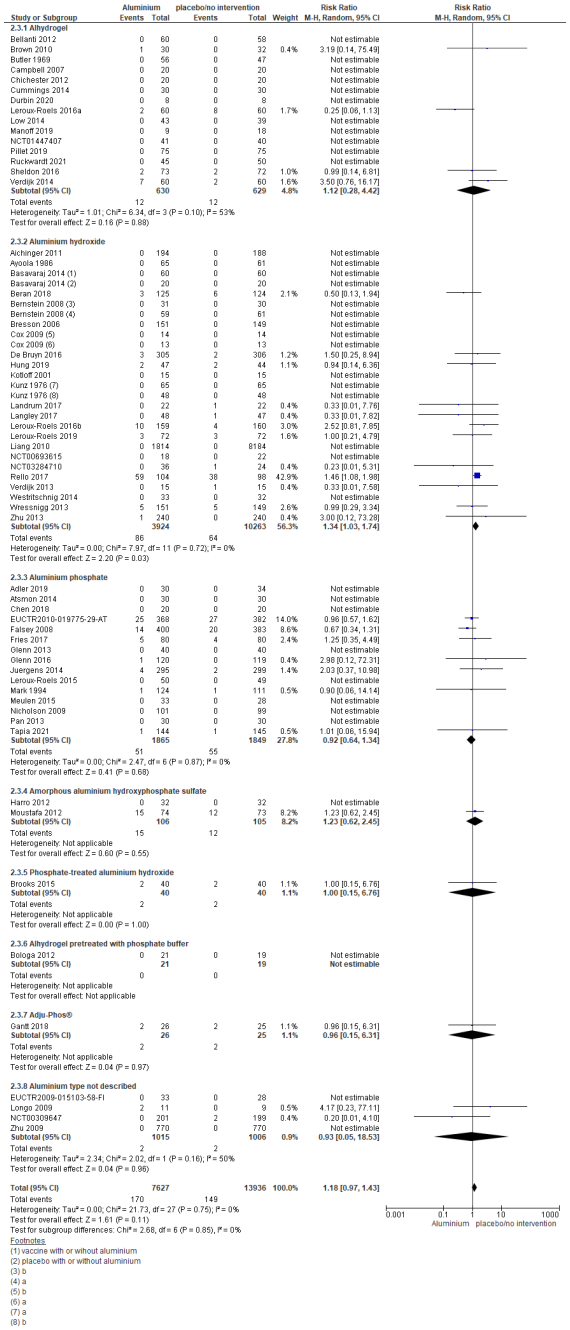
Supplementary Figure S4. Trial Sequential Analysis using random-effects model of the effects of aluminium adjuvants versus placebo or no intervention on the proportion of participants with serious adverse events. The analysis was based on a proportion in the control group (Pc) of 1.0 %, a relative risk reduction (or increase)(RRR) of 20%, an alpha of 2.5%, a beta of 10%, and diversity of 0%. None of the trial sequential monitoring boundaries have been reached or surpassed.



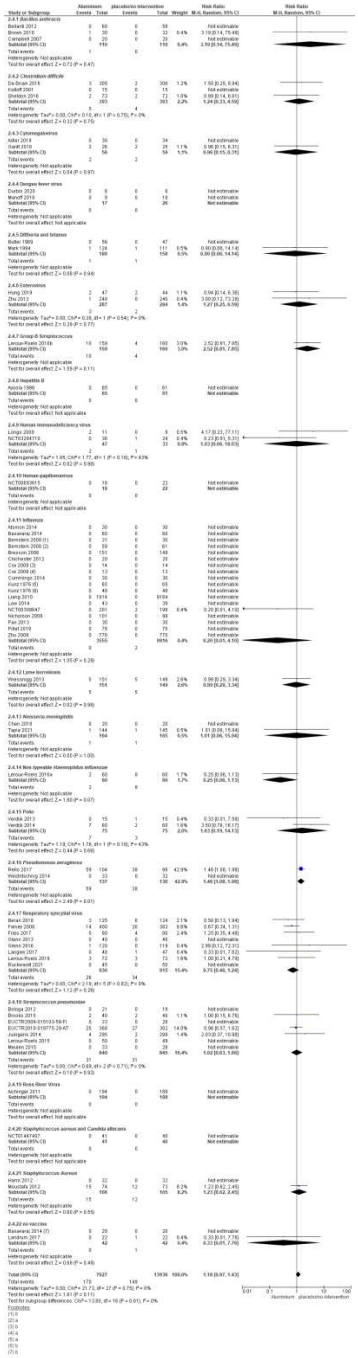
Supplementary Figure S5. Subgroup analysis on serious adverse events according to risk of bias. Random-effects model.



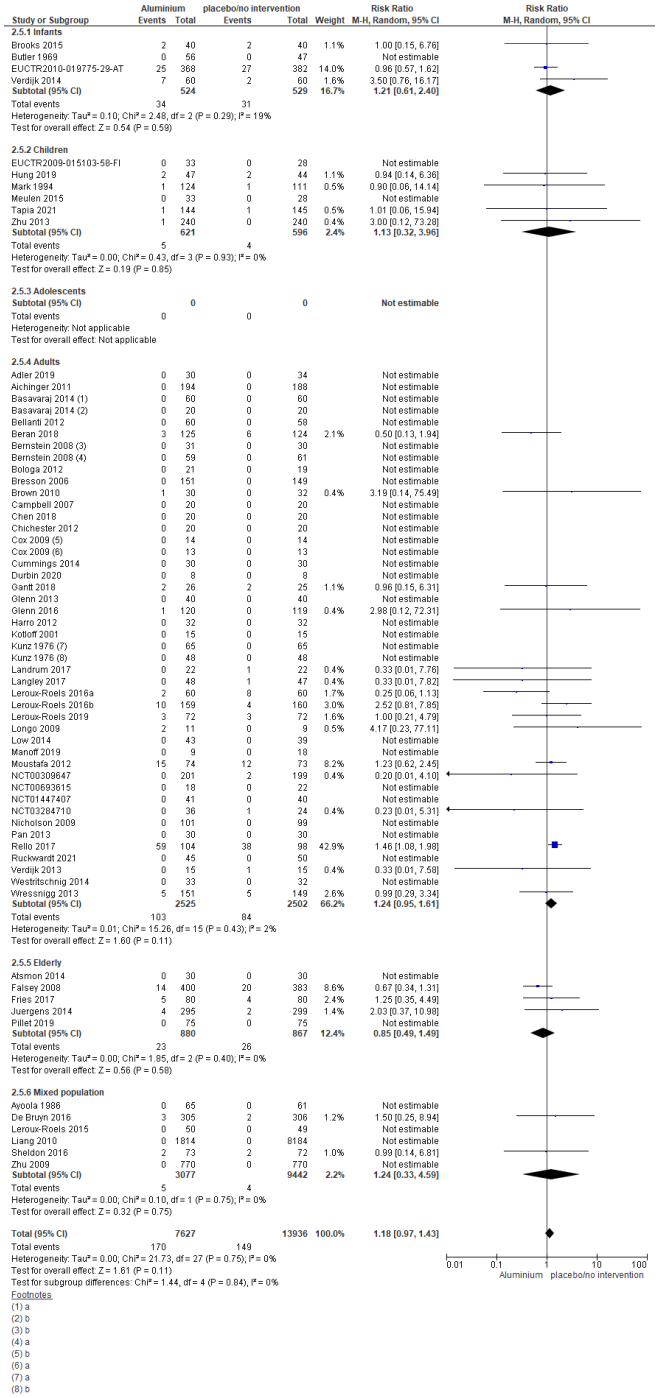
Supplementary Figure S6. Subgroup analysis of serious adverse events according to risk of vested interests. Random-effects model.



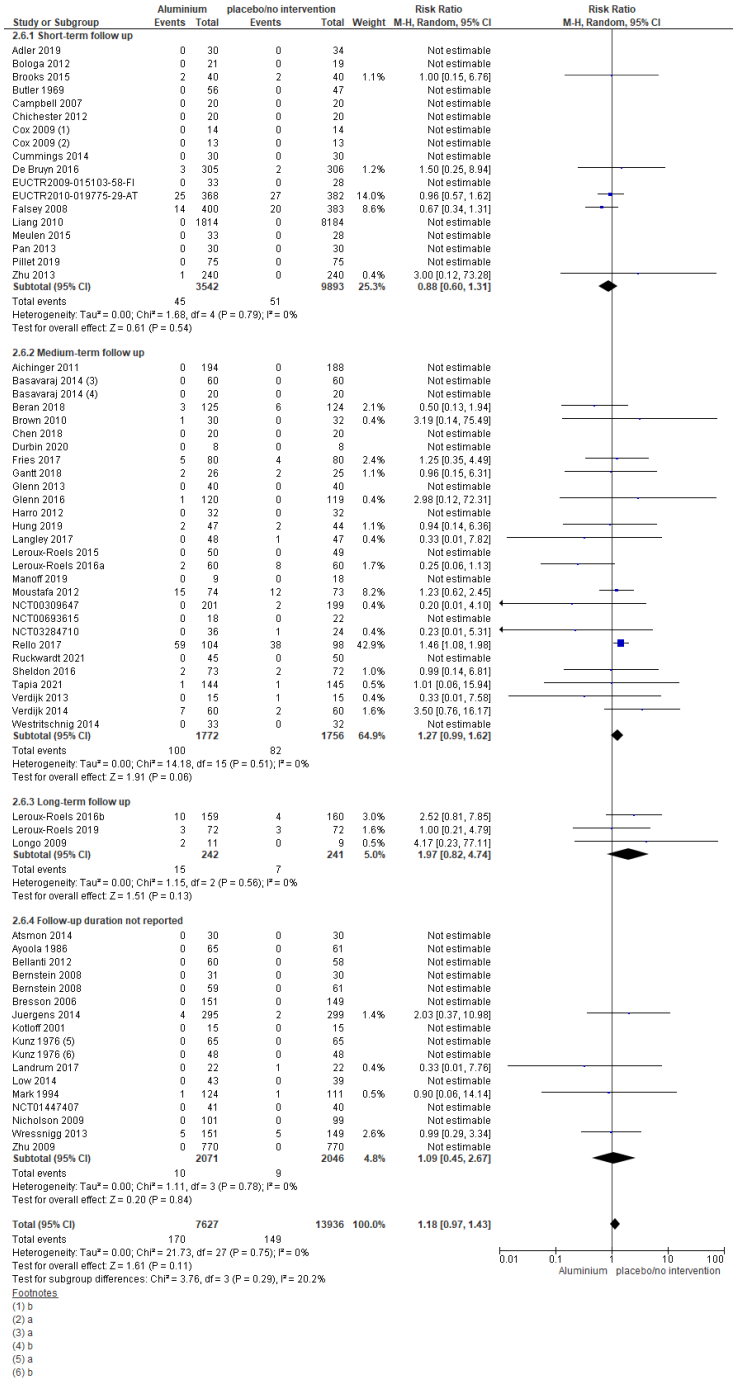
Supplementary Figure S7. Subgroup analysis of serious adverse events according to aluminium adjuvant type. Random-effects model.



Supplementary Figure S8. Subgroup analysis of serious adverse events according to type of vaccine. Random-effects model.

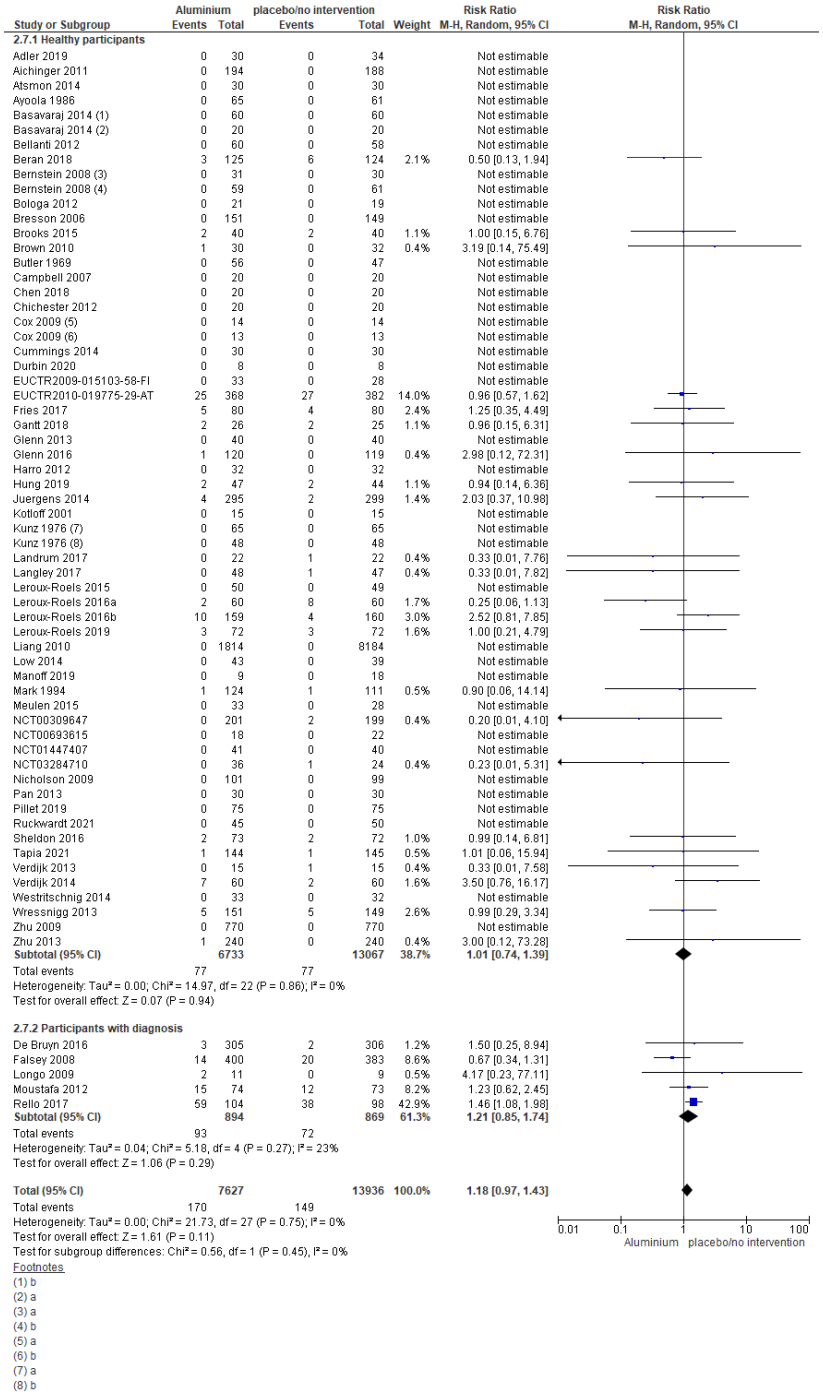


Supplementary Figure S9. Subgroup analysis of serious adverse events according to age groups of the participants.

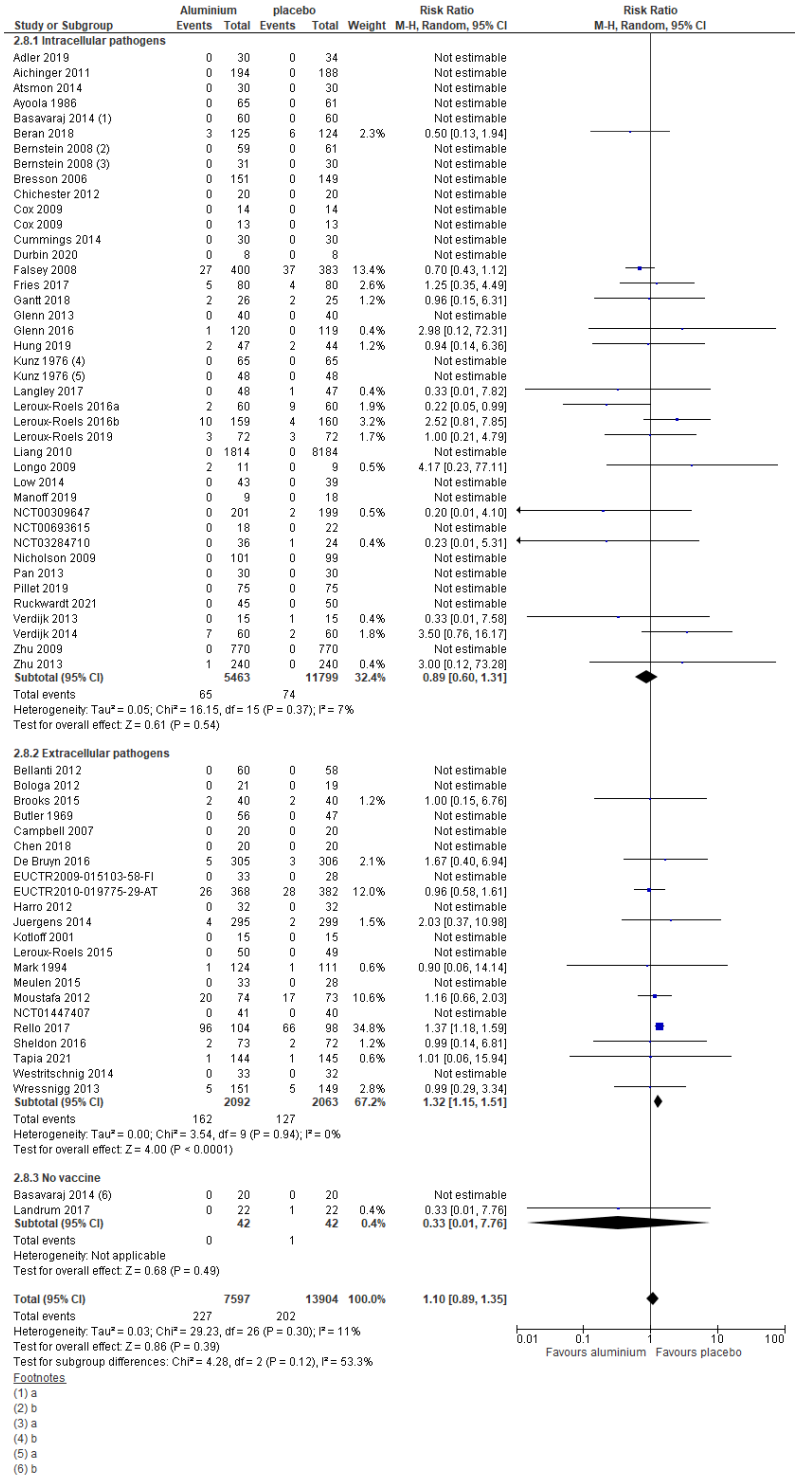


Supplementary Figure S10. Subgroup analysis of serious adverse events according to duration of follow up.

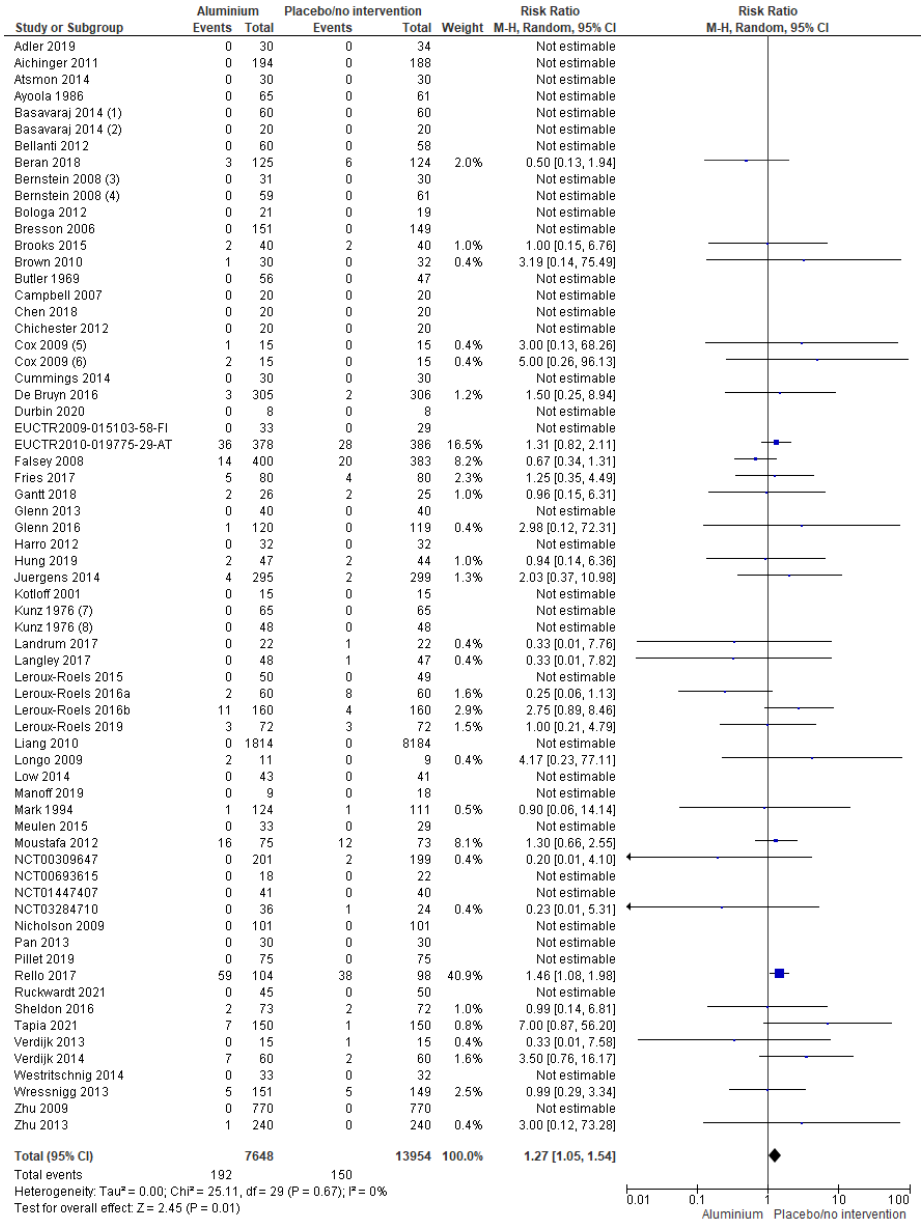
Random-effects model.



Supplementary Figure S11. Subgroup analysis of serious adverse events according to health status of the participants at entry into the trials. Random-effects model.

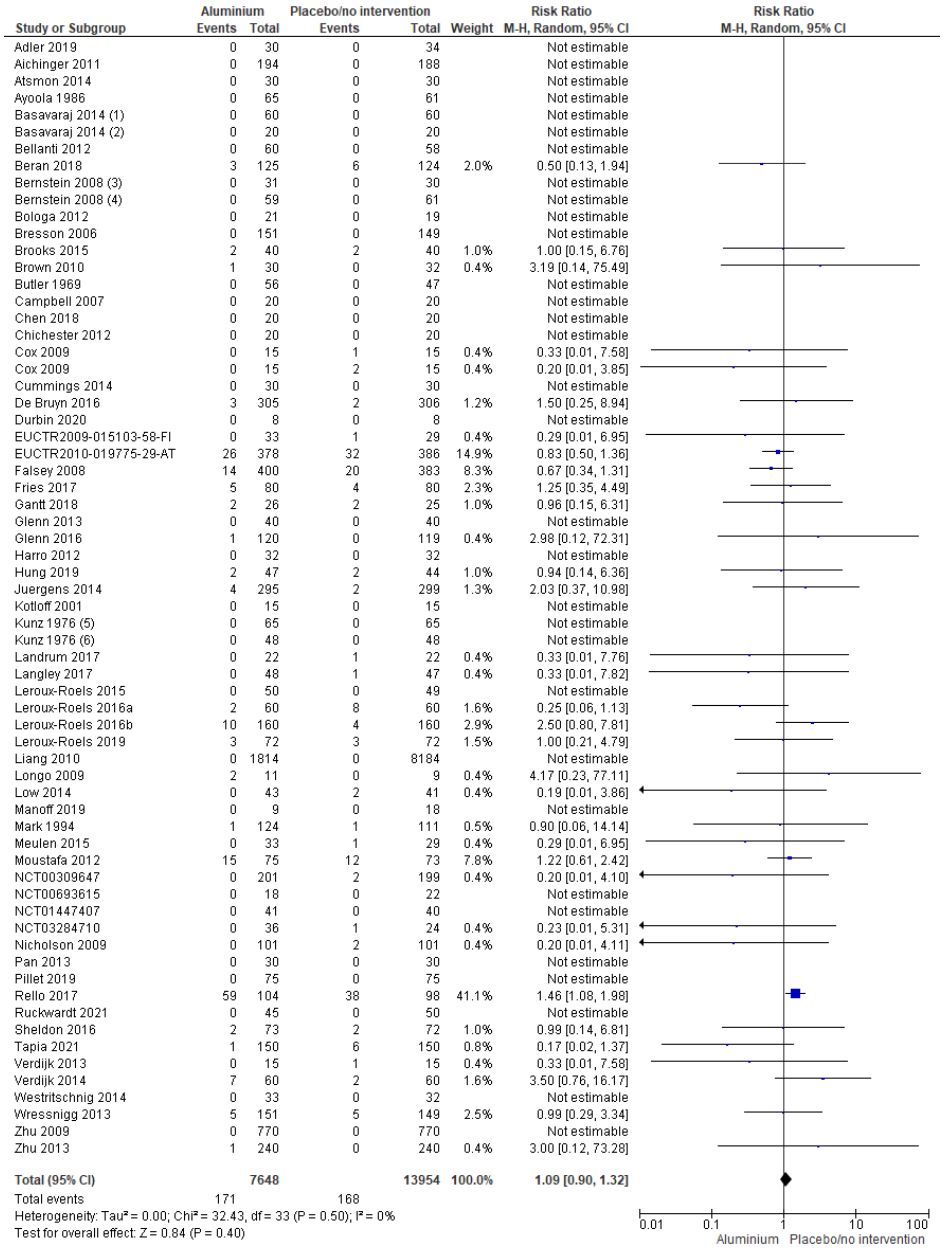


Supplementary Figure S12. Subgroup analysis of serious adverse events according to pathogen type the vaccine is directed against. Random-effects model.



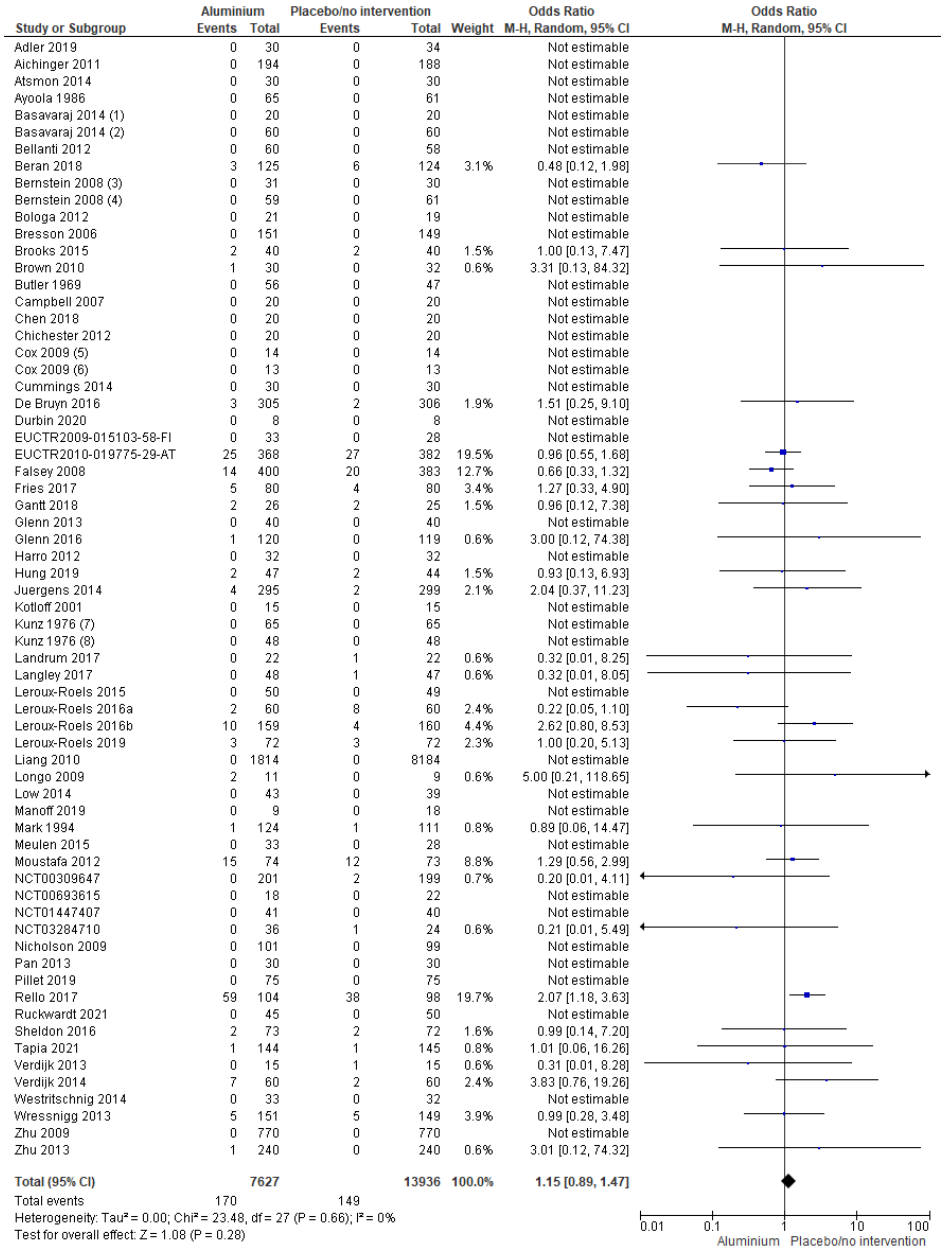
Supplementary Figure S13. Forest plot of 'worst best' case scenario sensitivity analysis of serious adverse events. Random-effects model.

We assessed the potential impact of missing data in the 'worst best' case scenario: here we assumed that all participants lost to follow-up in the experimental group had a serious adverse event; and all those with missing outcomes in the control group had no serious adverse events.



Supplementary Figure S14. Forest plot of 'best worst' case scenario sensitivity analysis of serious adverse events. Random-effects model.

We assessed the potential impact of missing data in the 'best worst' case scenario: here we assumed that all participants lost to follow-up in the control group had a serious adverse event; and all those with missing outcomes in the experimental group had no serious adverse events.



Supplementary Figure S15. Meta-analysis of serious adverse events using odds ratio. Random-effects model.

Supplementary Analysis S1.

Individual serious adverse events analysis

Meta-analysis showed no evidence of a difference between aluminium adjuvants versus control when assessing individual serious adverse events:

pneumonia (RR 0.59, 95% CI 0.16 to 2.12; trials = 6);

severe pulmonary embolism (RR 0.33, 95% CI 0.03 to 3.14; trials = 2);

gastroenteritis (RR 1.23, 95% CI 0.75 to 2.02; trials = 9);

pancreatitis (RR 0.99, 95% CI 0.10 to 9.44; trials = 2);

diarrhea (RR 0.82, 95% CI 0.17 to 3.86; trials = 4);

breast cancer (RR 0.33, 95% CI 0.05 to 2.08; trials = 3);

ligament rupture (RR 0.33, 95% CI 0.03 to 3.18; trials = 2);

depression (RR 0.33, 95% CI 0.03 to 3.08; trials = 2);

otitis (RR 0.22, 95% CI 0.03 to 1.92; trials = 2);

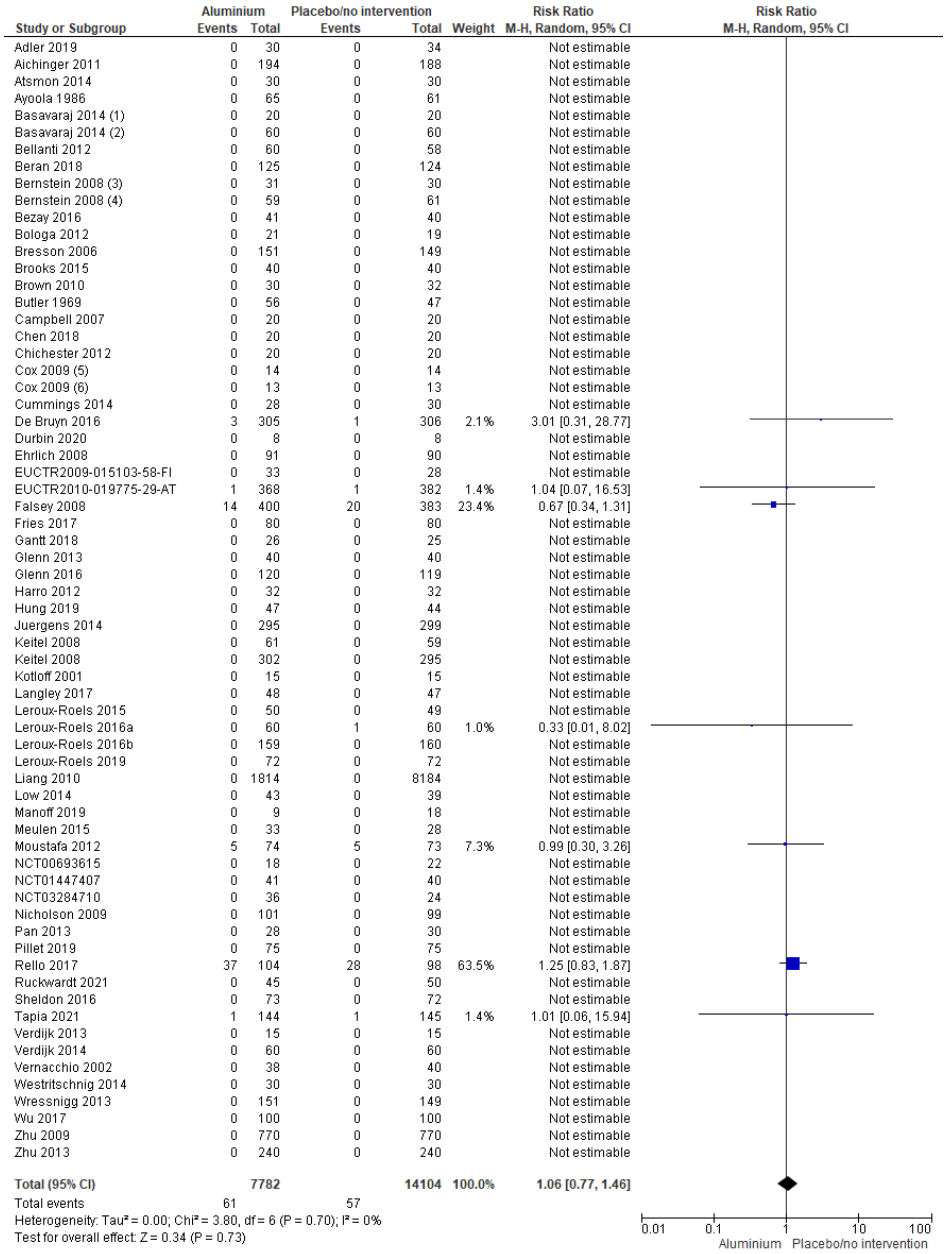
coronary artery disease (RR 2.98, 95% CI 0.31 to 28.28; trials = 2);

cardiac arrest (RR 1.61, 95% CI 0.08 to 31.86; trials = 2);

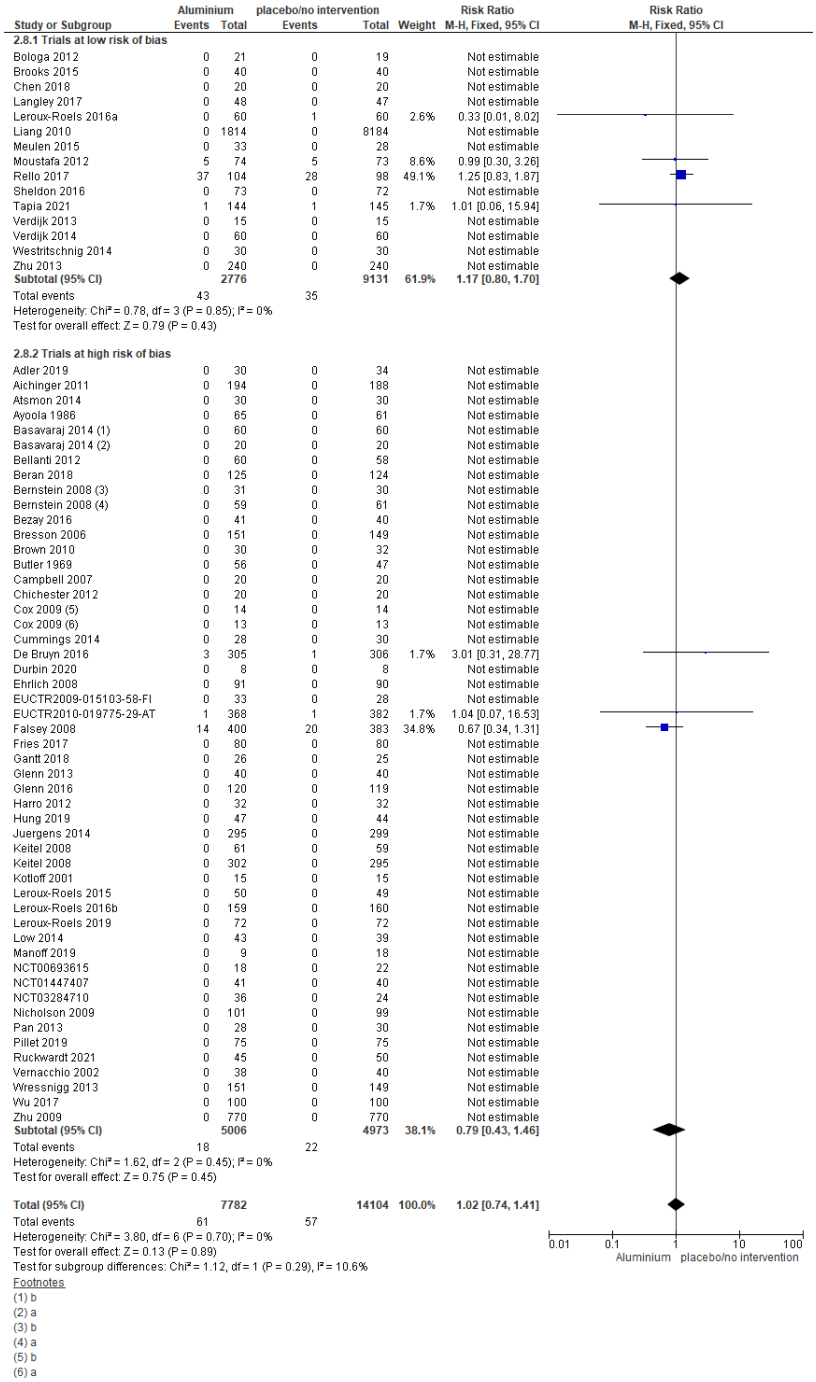
sepsis (RR 0.94, 95% CI 0.14 to 6.34; trials = 2).

Serious adverse events reported in the aluminum-adjuvanted group versus the placebo group		
Serious adverse event	**Aluminium event/total (study ID)**	**Placebo event/total (study ID)**
Astrocytoma	0/22 (Landrum2017)	1/22 (Landrum2017)
Congenital laryngomalacia	1/40 (Brooks 2015)	0/40 (Brooks 2015)
Acute appendicitis	1/30 (Glenn 2016)	0/29 (Glenn 2016)
Traumatic left knee dislocation	0/16 (Langley 2017)	1/16 (Langley 2017)
Tonsillitis	1/26 (Hung 2019)	1/24 (Hung 2019)
Cardiac failure	1/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Pulmonary hemorrhage	1/104 (Rello 2017)	0/98 (Rello 2017)
Shock	1/104 (Rello 2017)	0/98 (Rello 2017)
Prostate cancer	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Myocardial infarction	0/15 (Verdrijk 2013	1/15 (Verdrijk 2013
Moderate infective bursitis	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Severe cerebral metastasis and bleed	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Lung cancer	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Cubital tunnel syndrome	0/151 (Wressnigg 2013)	1/149 (Wressnigg 2013)
Intravertebral disc protusion	0/151 (Wressnigg 2013)	1/149 (Wressnigg 2013)
Pulmonary embolism and deep vein	1/151 (Wressnigg 2013)	0/149 (Wressnigg 2013)
Bronchopneumonia	0/26 (Hung 2019)	1/24 (Hung 2019)
Viral encephalitis	1/120 (Zhu 2013)	0/120 (Zhu 2013)
Benign prostatic hyperplasia	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Elective abortion	0/151 (Wressnigg 2013)	1/149 (Wressnigg 2013)
Musculoskeletal injuries	0/125 (Beran 2018)	1/124 (Beran 2018)
Spontaneous abortion	1/125 (Beran 2018)	0/124 (Beran 2018)
Transient ischemic attack	0/125 (Beran 2018)	1/124 (Beran 2018)
Umbilical hernia	1/125 (Beran 2018)	0/124 (Beran 2018)
Virally induced asthma	1/125 (Beran 2018)	0/124 (Beran 2018)
Colitis	1/125 (Beran 2018)	0/124 (Beran 2018)
Incisional hernia	1/125 (Beran 2018)	0/124 (Beran 2018)
Weber b fracture right ankle	1/125 (Beran 2018)	0/124 (Beran 2018)
Tibia fracture	1/125 (Beran 2018)	0/124 (Beran 2018)
Erysipelas cheek	1/125 (Beran 2018)	0/124 (Beran 2018)
Arrhythmia	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Cardiogenic Shock	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Gastric ulcer haemorrhage	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Gangrene	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Osteomyelitis bacteria	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Staphylococcal infection	1/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Ankle fracture	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Arteriovenous fistula occlusion	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Joint injury	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Diabetic ketoacidosis	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Azotaemia	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
COPD	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Pulmonary oedema	0/74 (Moustafa 2012)	1/73 (Moustafa 2012)
Thrombosis	1/74 (Moustafa 2012)	0/73 (Moustafa 2012)
Roseola infantum	0/26 (Hung 2019)	1/24 (Hung 2019)
Dehydration	0/26 (Hung 2019)	2/24 (Hung 2019)
Hand, foot, mouth disease	1/26 (Hung 2019)	0/24 (Hung 2019)
Clostridium difficile infection	1/26 (Hung 2019)	0/24 (Hung 2019)
Acute pyelonephritis	0/72 (Lerouz-Roels 2019)	1/72 (Lerouz-Roels 2019)
Cartilage injury	1/72 (Lerouz-Roels 2019)	0/72 (Lerouz-Roels 2019)
Head injury	0/72 (Lerouz-Roels 2019)	1/72 (Lerouz-Roels 2019)
Anaphylactic shock	1/72 (Lerouz-Roels 2019)	0/72 (Lerouz-Roels 2019)
Transient paralysis	1/25 (Bellino 2009)	0/22 (Bellino 2009)
Lipothymia-nausea	1/25 (Bellino 2009)	0/22 (Bellino 2009)
Laceration	0/26 (Gantt 2018)	1/25 (Gantt 2018)
Infections	1/124 (Mark 1994)	1/111 (Mark 1994)
Deep vein thrombosis	1/26 (Gantt 2018)	0/25 (Gantt 2018)

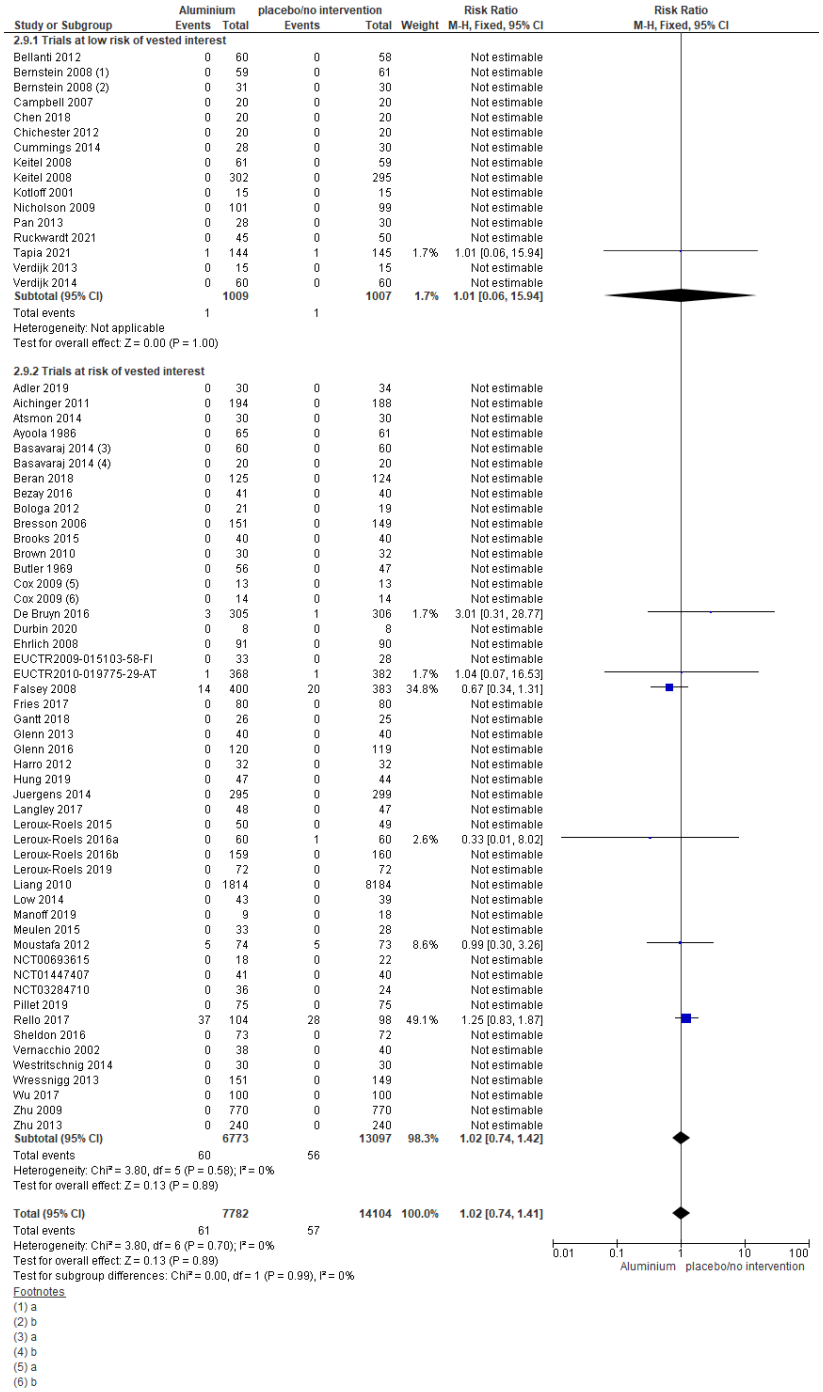
Supplementary Table S2. List of individual serious adverse events reported in only one trial. It was our aim to examine if the serious adverse events in the aluminium-adjuvanted groups of the included trials resembled the syndromes described in the literature related to aluminium toxicity. All included trials reporting serious adverse events were screened for detailed information about the events. However, the reporting was too sketchy to use the information for comparison with the syndromes. In general, the types of serious adverse events were not specified in the trials, and it was impossible to assess the total number of serious adverse events for each trial participant. We can therefore not evaluate whether the serious adverse events are related to what is described as 'aluminium toxicity' in the literature.



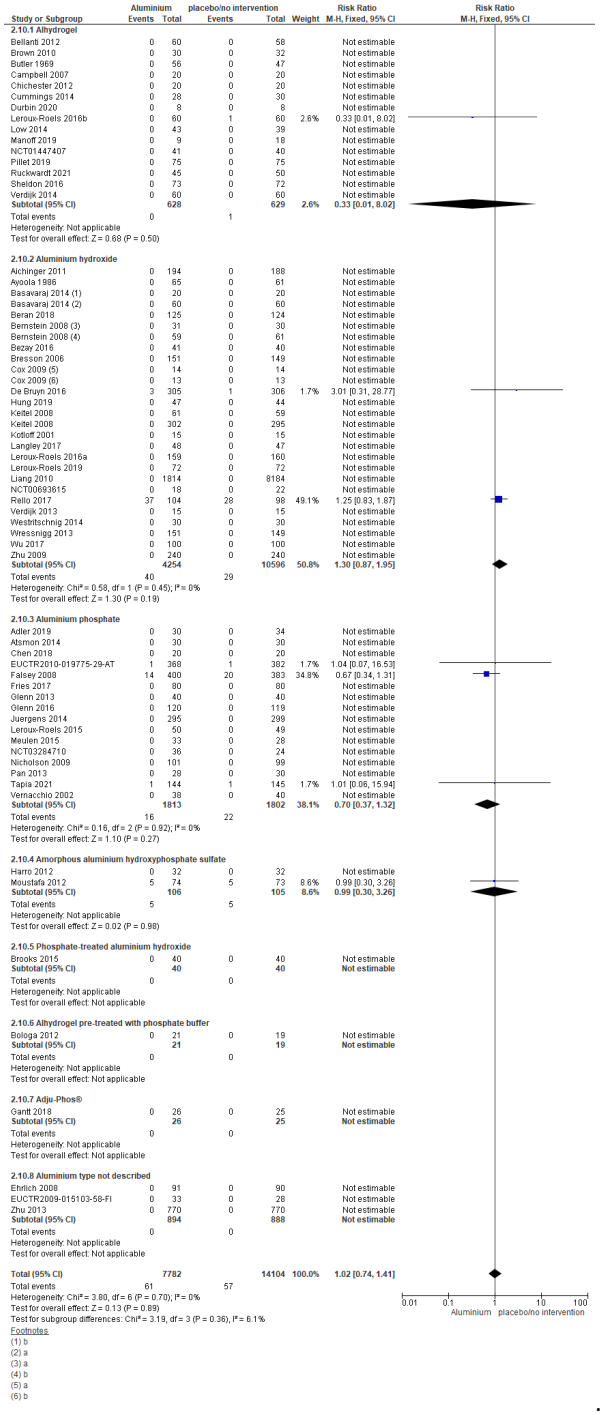
Supplementary Figure S16. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on all-cause mortality. Random-effects model.



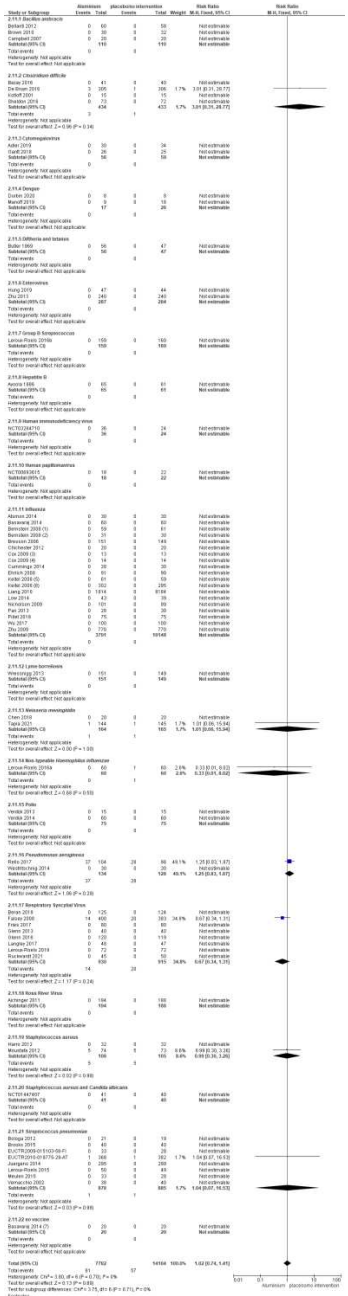
Supplementary Figure S17. Subgroup analysis of all-cause mortality according to risk of bias. Fixed-effect model.



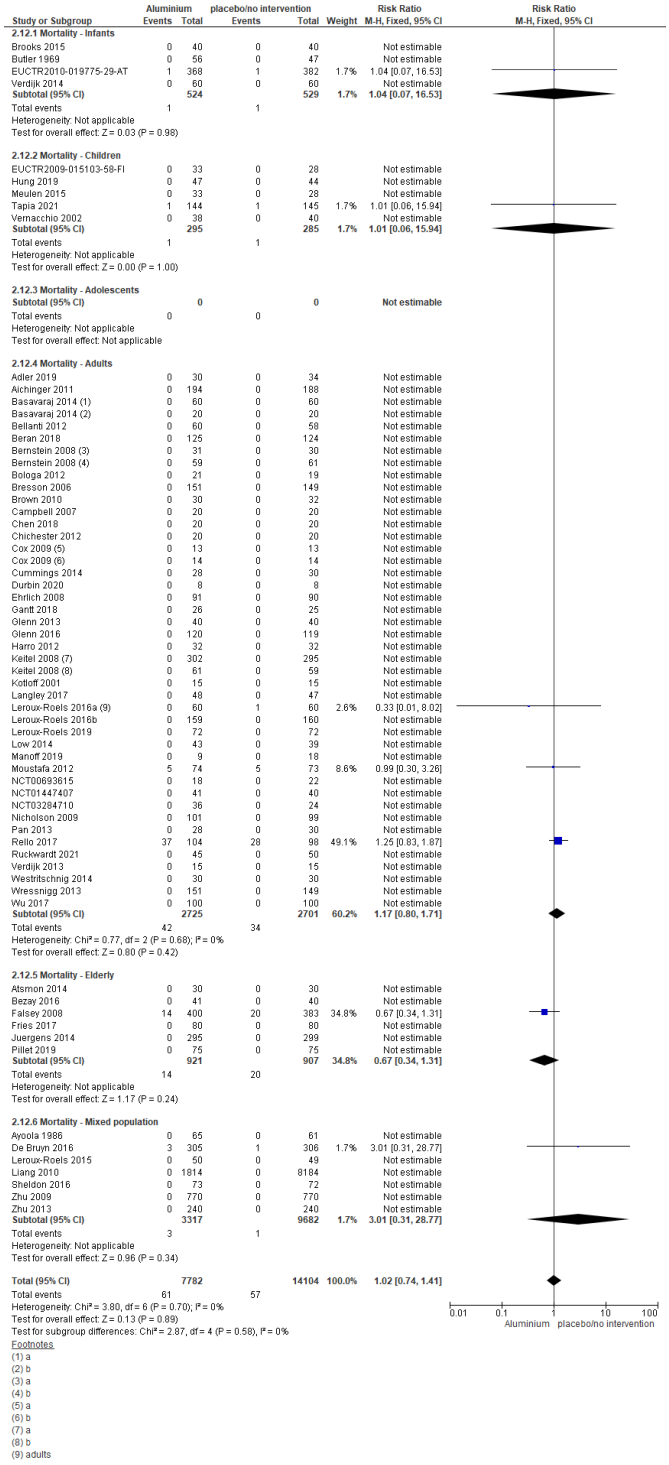
Supplementary Figure S18. Subgroup analysis of all-cause mortality according to risk of vested interests. Fixed-effect model.



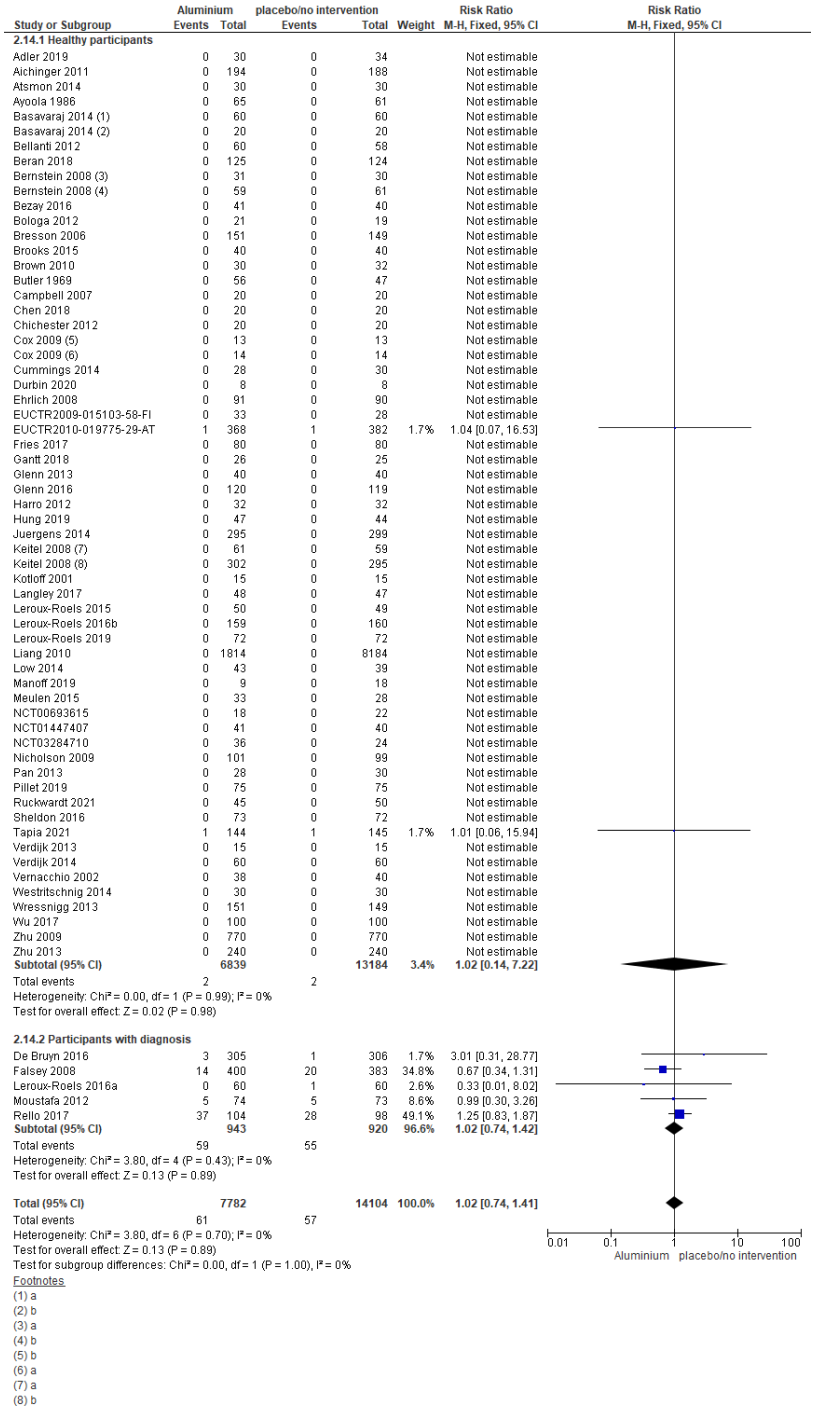
Supplementary Figure S19. Subgroup analysis of all-cause mortality according to aluminium adjuvant type. Fixed-effect model.



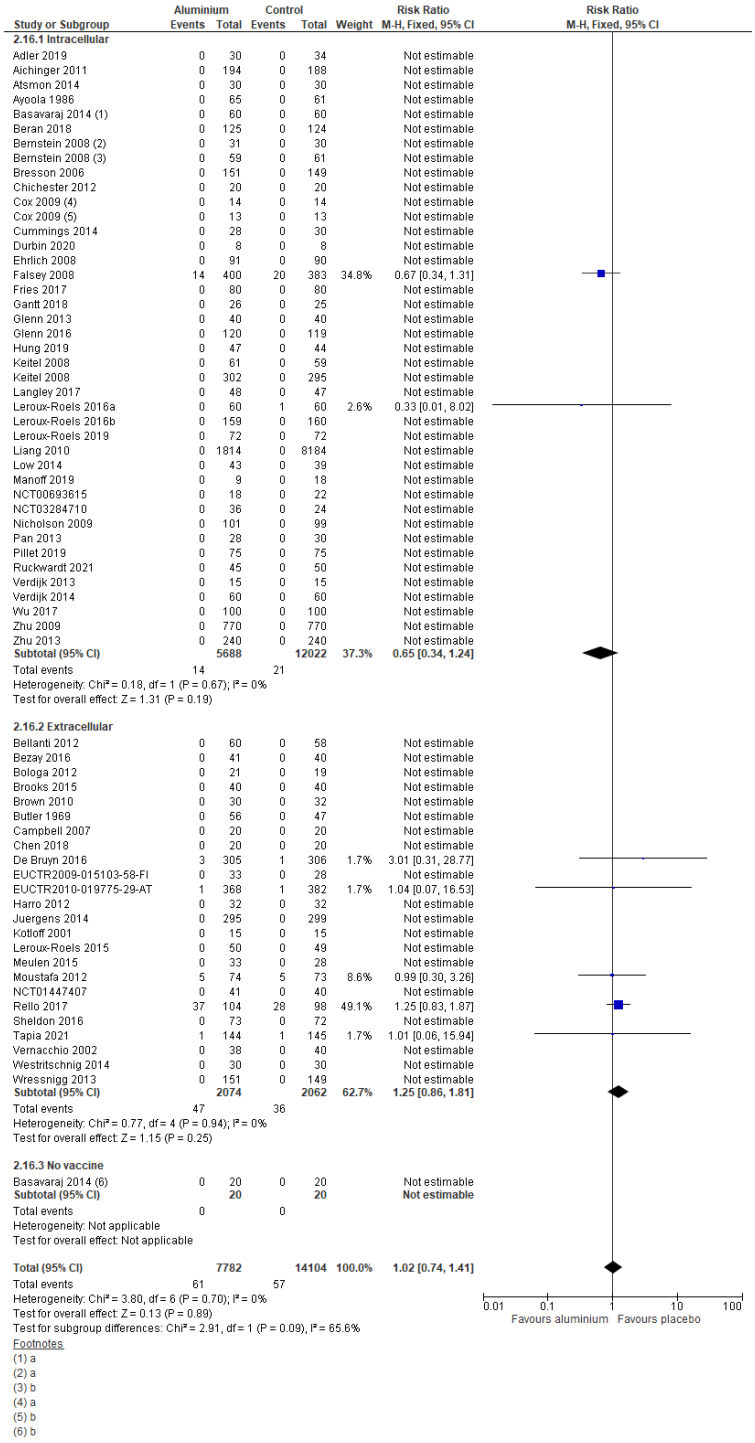
Supplementary Figure S20. Subgroup analysis of all-cause mortality according to vaccine type. Fixed-effect model.



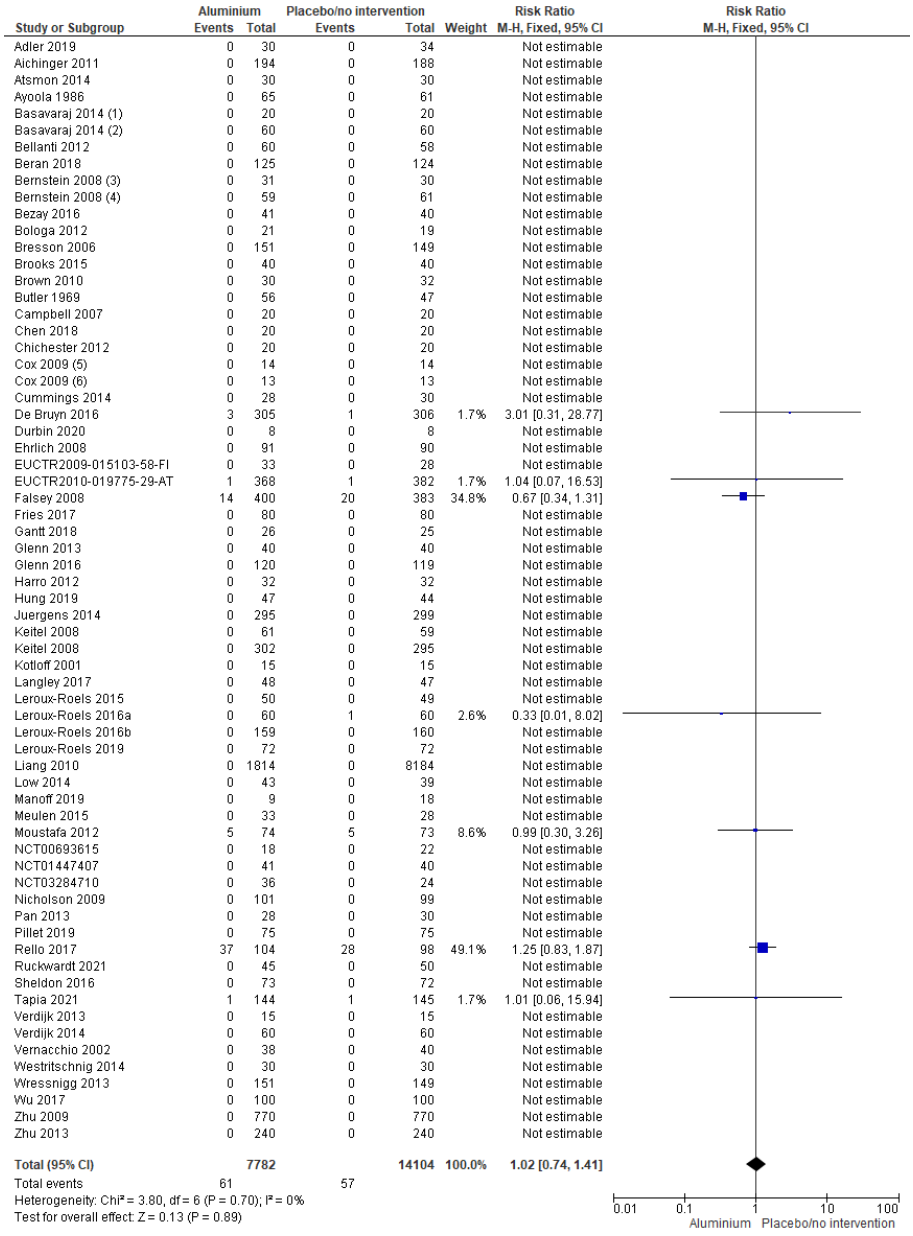
Supplementary Figure S21. Subgroup analysis of all-cause mortality according to age groups of the participants. Fixed-effect model.



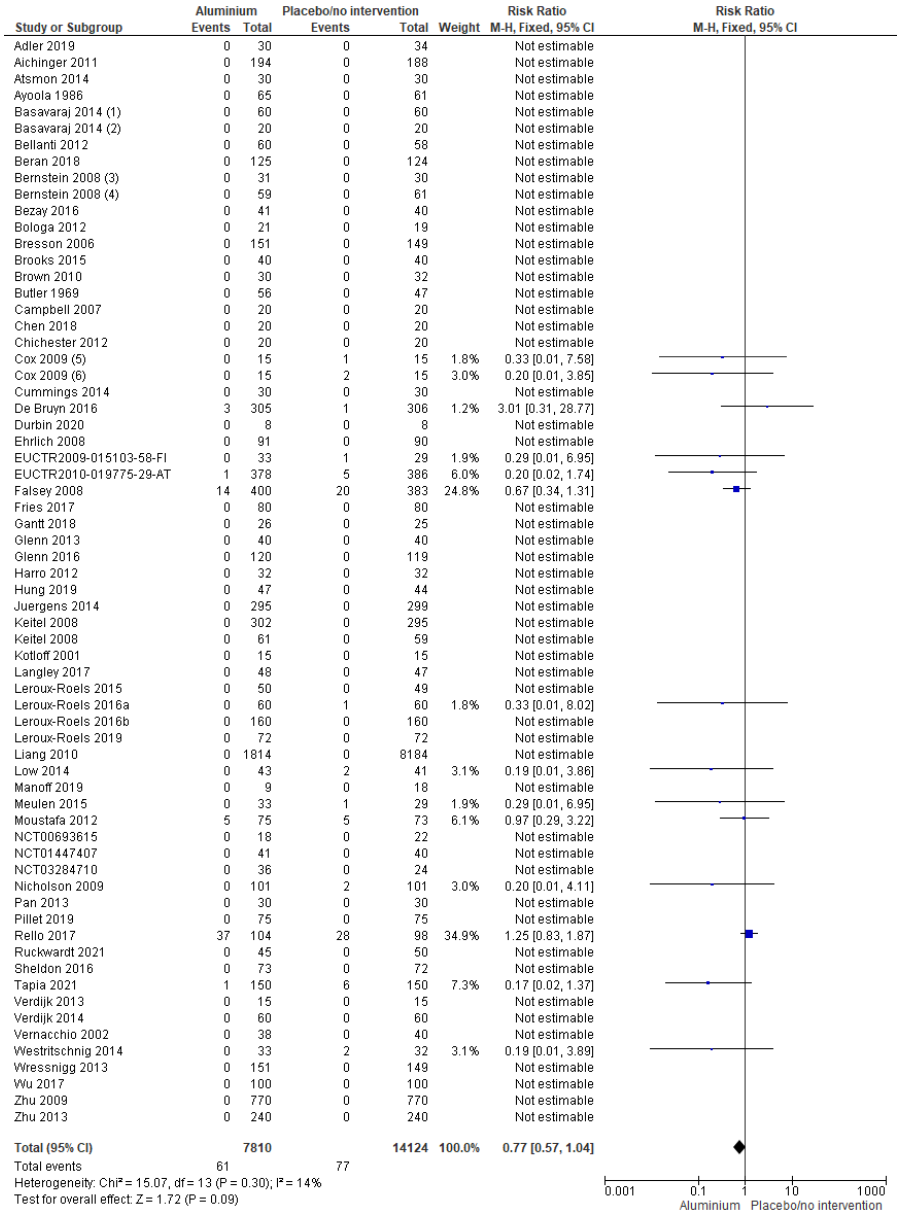
Supplementary Figure S22. Subgroup analysis of all-cause mortality according to health status of the participants at entry into the trial. Fixed-effect model.



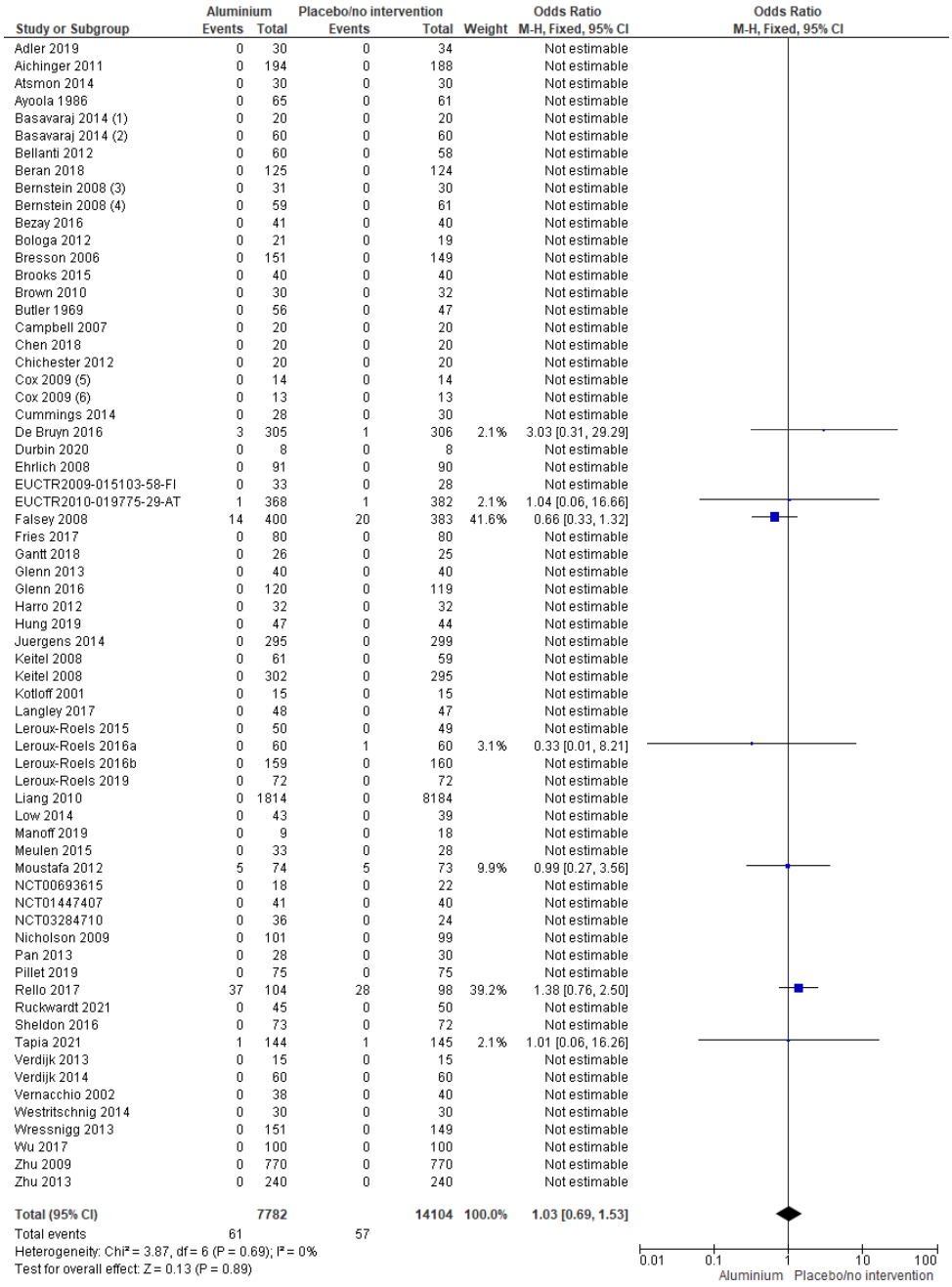
Supplementary Figure S23. Subgroup analysis of all-cause mortality according to pathogen type the vaccine is directed against. Fixed-effect model.



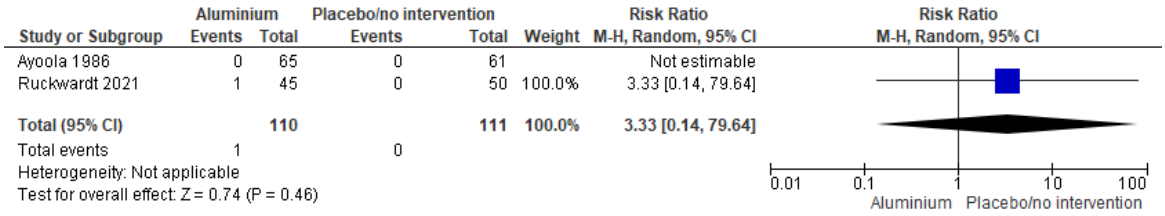
Supplementary Figure S24. Forest plot of 'worst best' case scenario sensitivity analysis of all-cause mortality. We assessed the potential impact of missing data in the 'worst best' case scenario: here we assumed that all participants lost to follow-up in the experimental group did not survive; and all those with missing outcomes in the control group did survive. Fixed-effect model.



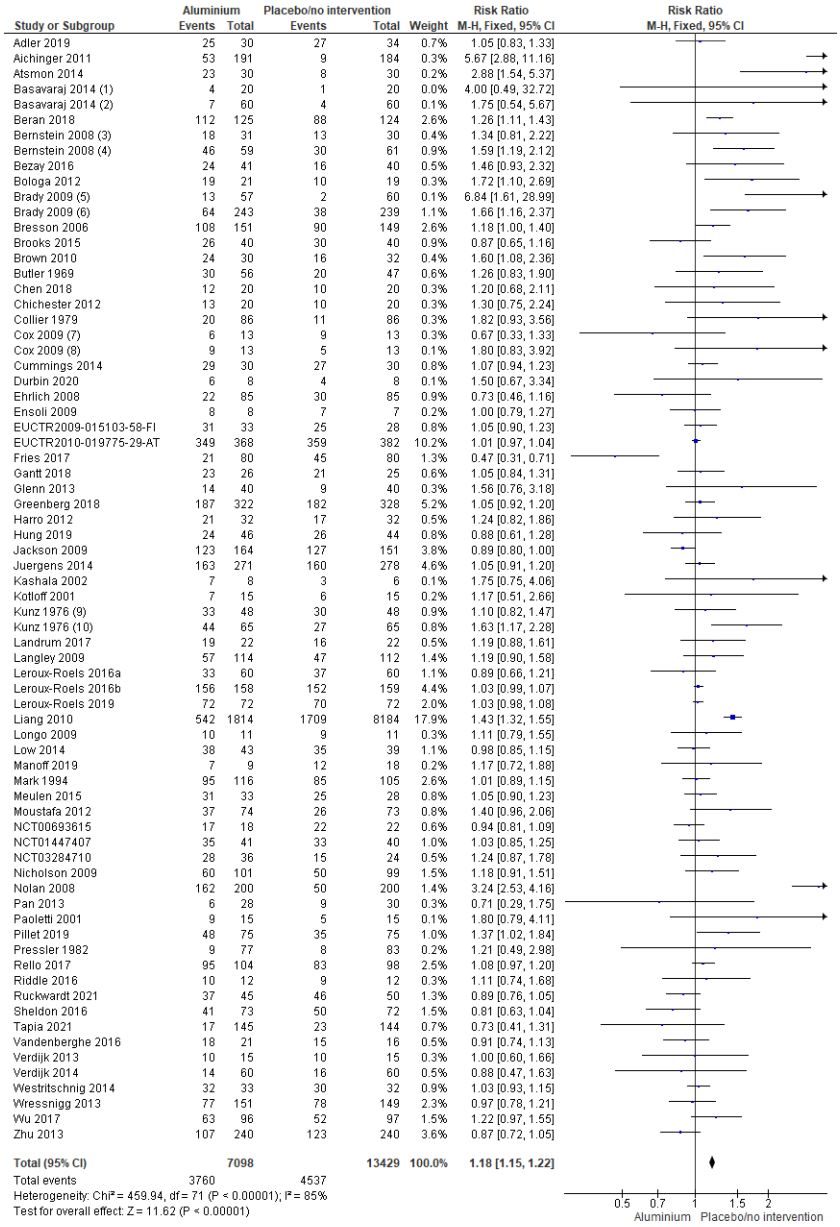
Supplementary Figure S25. Forest plot of 'best worst' case scenario sensitivity analysis of all-cause mortality. We assessed the potential impact of missing data in the 'best worst' case scenario: here we assumed that all participants lost to follow-up in the control group did not survive; and all those with missing outcomes in the experimental group did survive. Fixed-effect model.



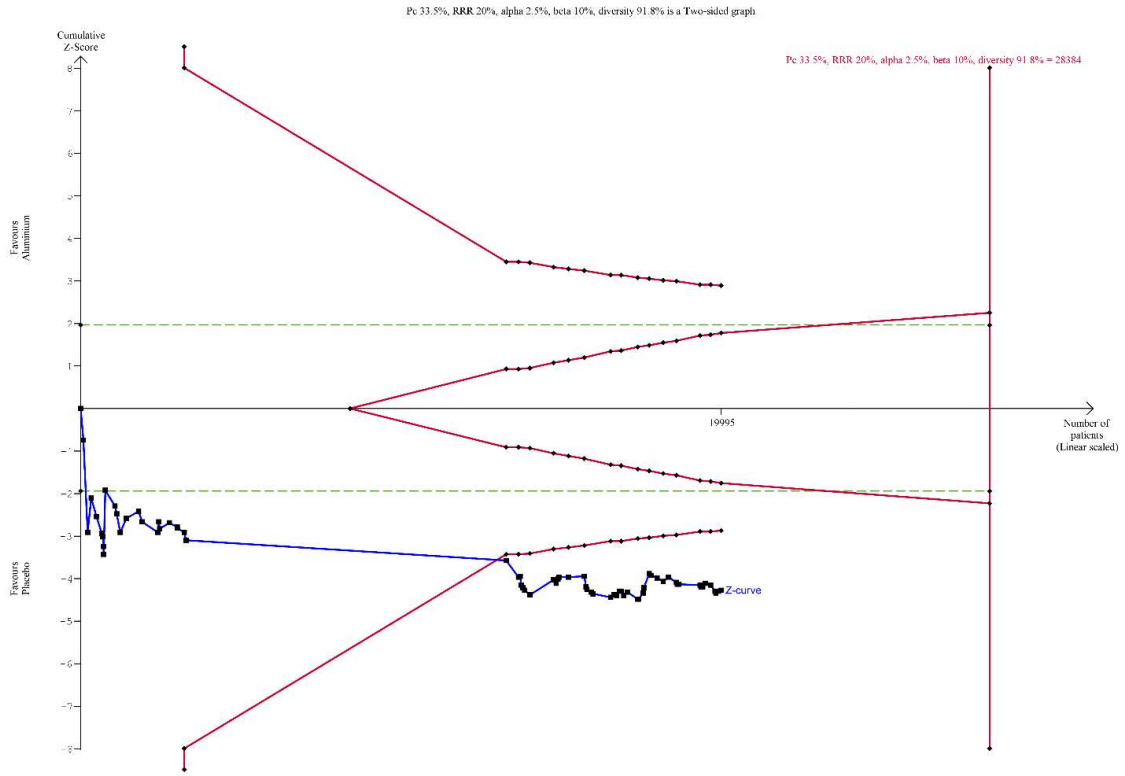
Supplementary Figure S26. Meta analysis of all-cause mortality using odds ratio. Fixed-effect model.



Supplementary Figure S27. Meta-analysis of the effect of aluminium adjuvants compared with placebo on the proportion of participants with disease being vaccinated against. Random-effects model.



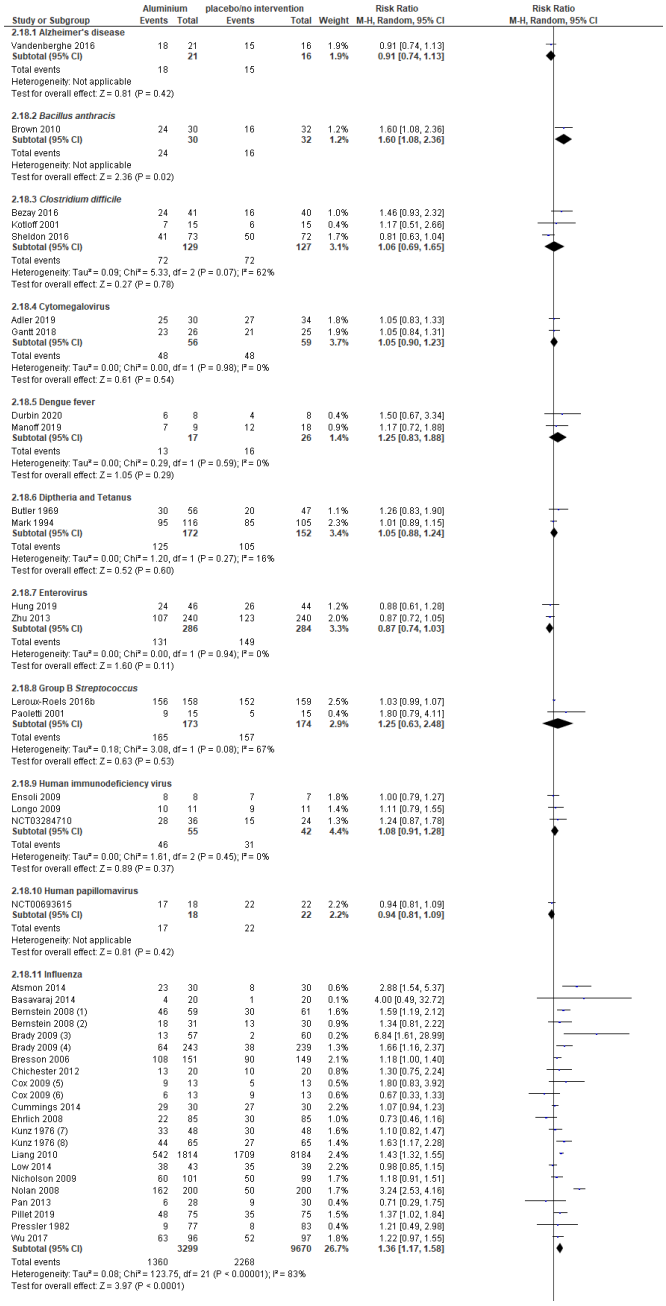
Supplementary Figure S28. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the proportion of participants with one or more non-serious adverse events. Fixed-effect model.



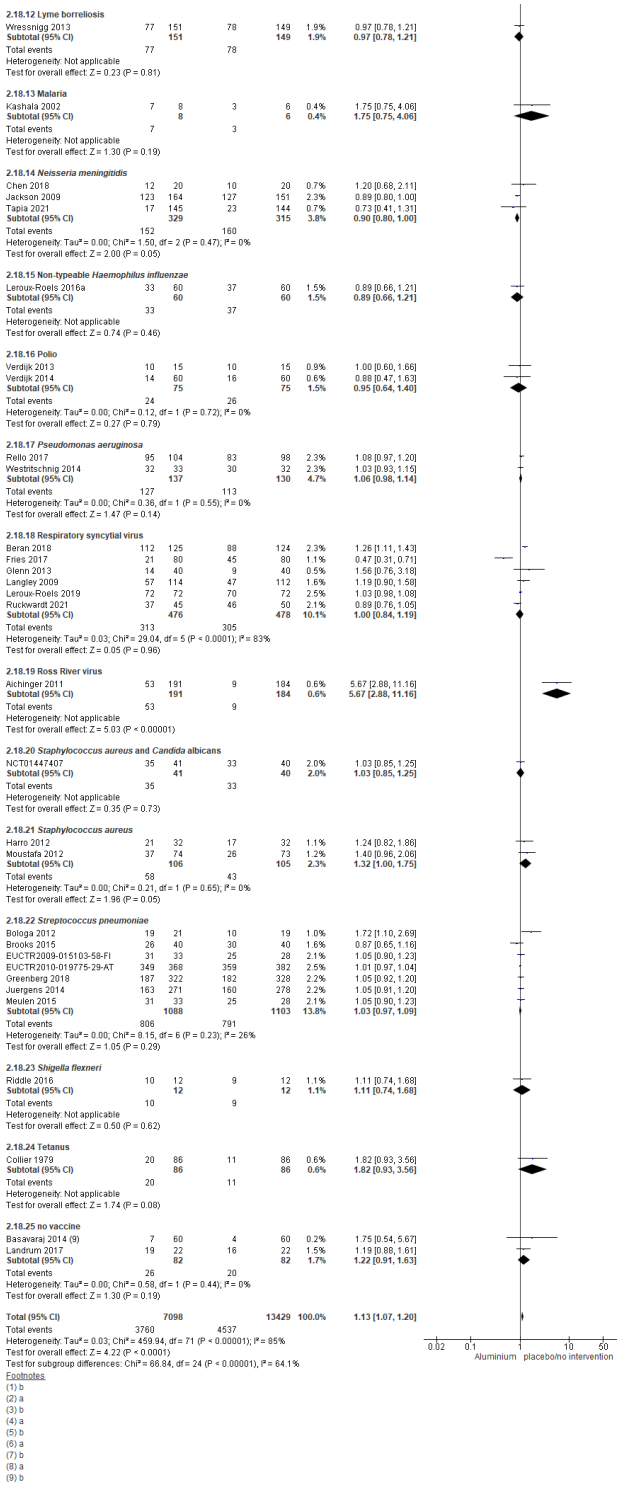
Supplementary Figure S29.

Trial Sequential Analysis using random-effects model of the effects of aluminium adjuvants versus placebo or no intervention on the proportion of participants with adverse events considered not-serious. The analysis was based on a proportion in the control group (Pc) of 33.8%, a relative risk reduction (or increase) (RRR) of 20%, an alpha of 2.5%, a beta of 10% and diversity of 92%. The graph shows that there was enough information to confirm that aluminium adjuvants compared with placebo or no intervention increases the risk of one or more non-serious adverse events by more than 20% relative risk increase.

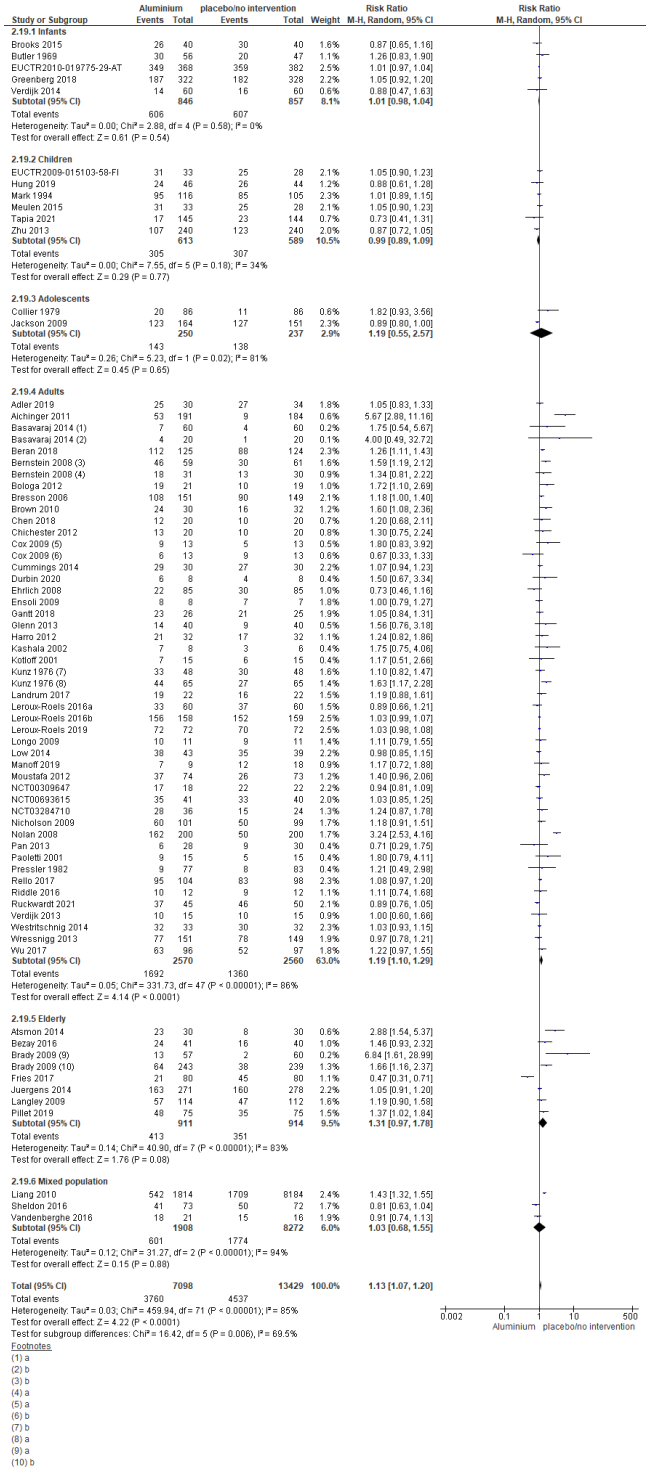
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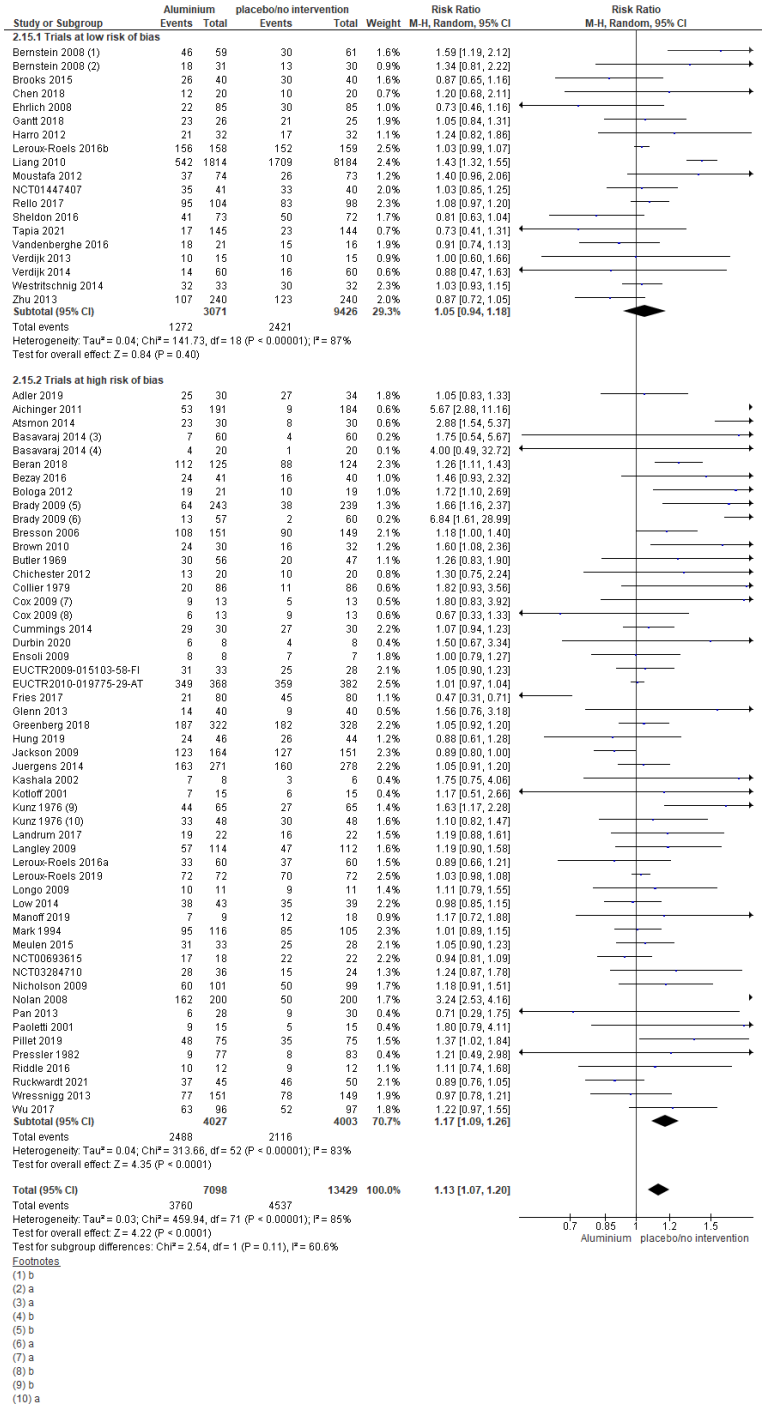
continues below



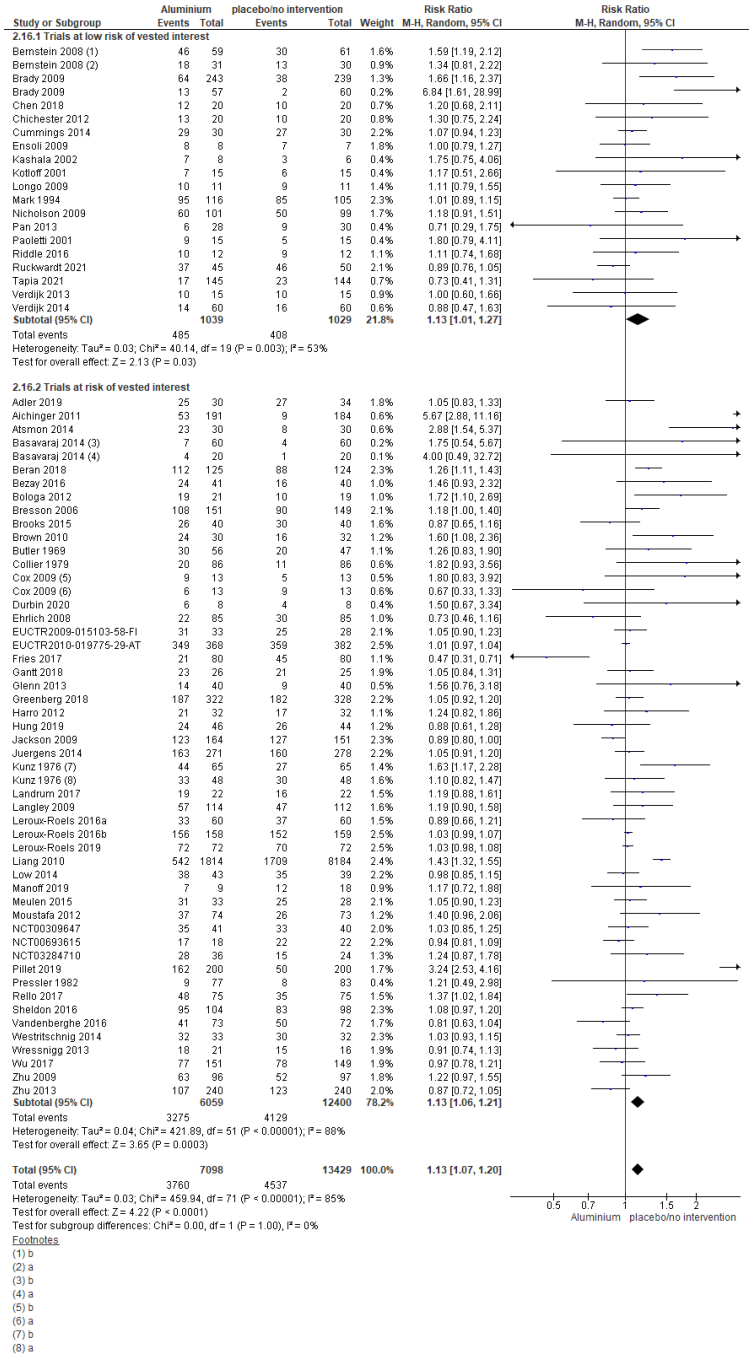
Supplementary Figure S30. Subgroup analyses of non-serious adverse events according to pathogen type the vaccine is directed against. Random-effects model.



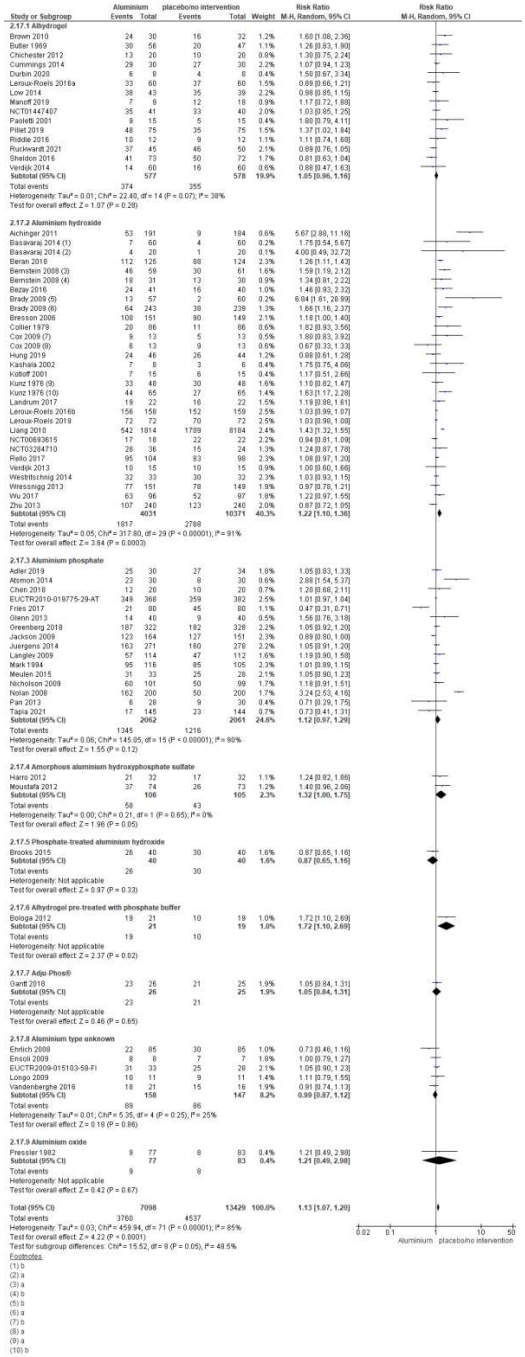
Supplementary Figure S31. Subgroup analyses of non-serious adverse events according to age groups of the participants. Random-effects model.



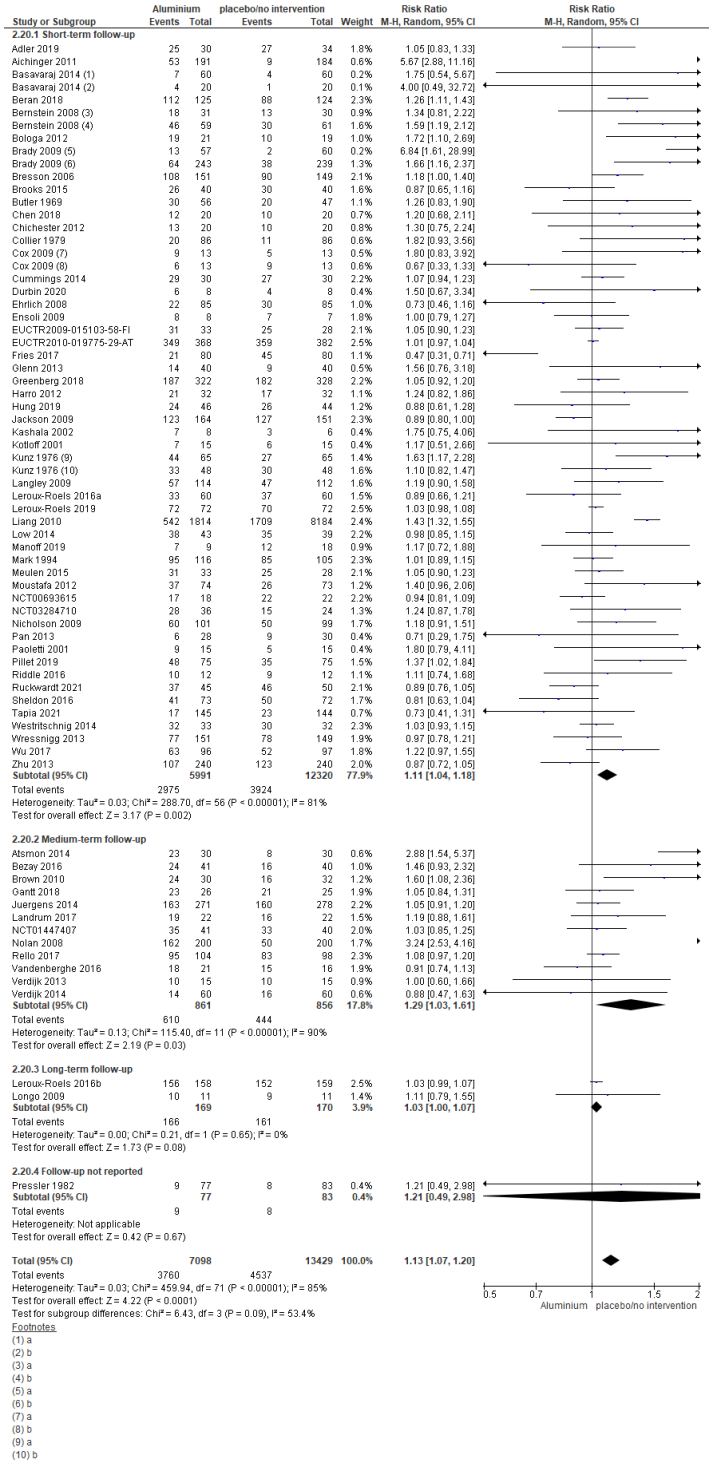
Supplementary Figure S32. Subgroup analyses of adverse events considered non-serious according to risk of bias. Random-effects model.



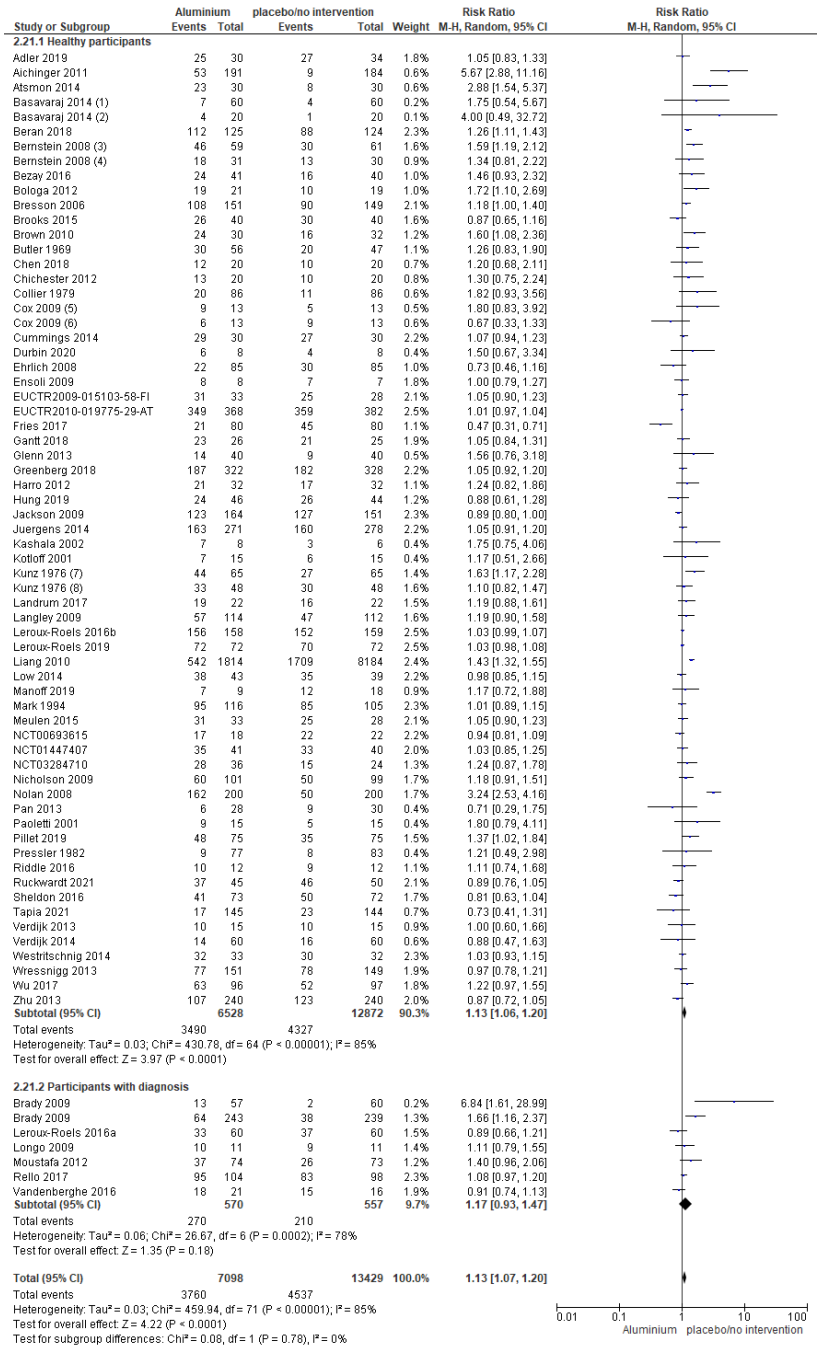
Supplementary Figure S33. Subgroup analyses of adverse events considered non-serious according to vested interests. Random-effects model.



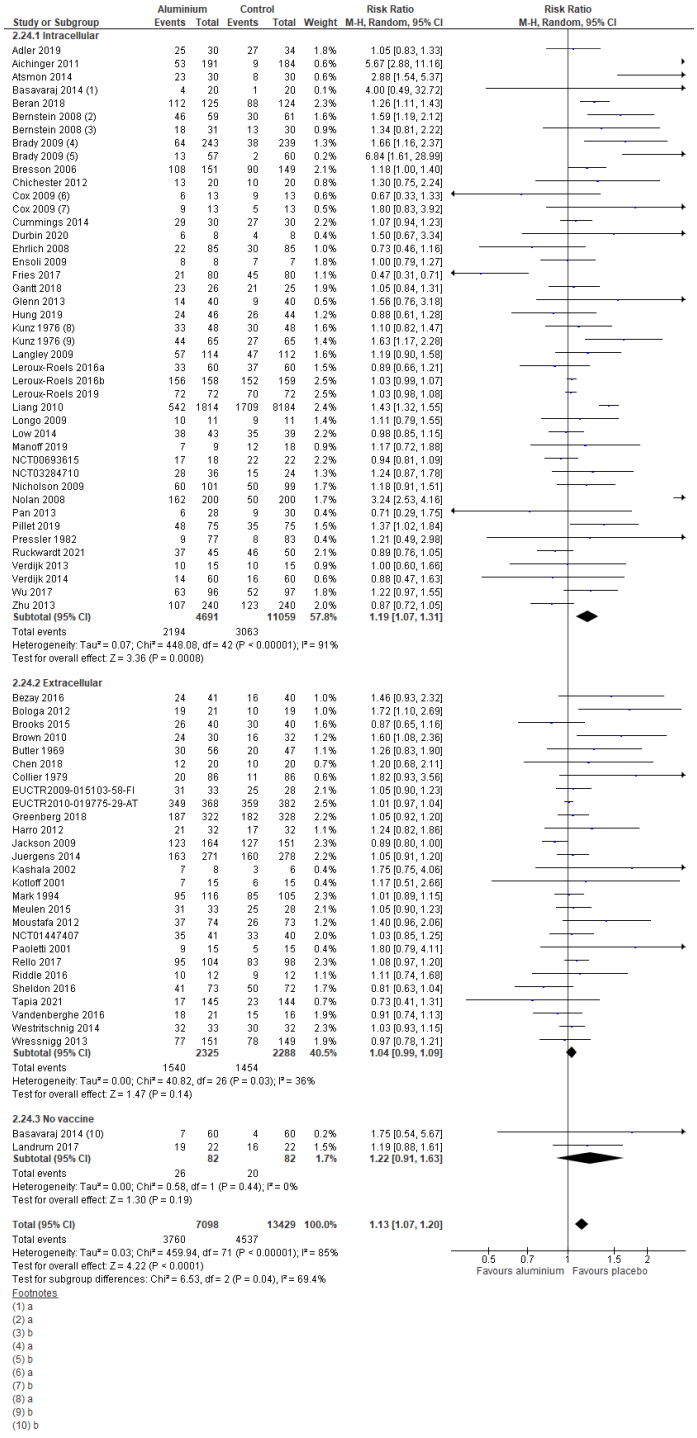
Supplementary Figure S34. Subgroup analyses of adverse events considered non-serious according to aluminium adjuvant type. Random-effects model.



Supplementary Figure S35. Subgroup analyses of adverse events considered non-serious according to duration of follow-up. Random-effects model.


Aluminium
placebo/no intervention
Risk Ratio
Risk Ratio
Study or Subgroup
Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
2.21.1 Healthy participants
Adler 2019 25 30 27 34 1.8% 1.05 [0.83, 1.33]
Aichinger 2011 53 191 9 184 0.6% 5.67 [2.88, 11.16]
Atsmon 2014 23 30 8 30 0.6% 2.88 [1.54, 5.37]
Basavaraj 2014 (1) 7 60 4 60 0.2% 1.75 [0.54, 5.67]
Basavaraj 2014 (2) 4 20 1 20 0.1% 4.00 [0.49, 32.72]
Beran 2018 112 125 88 124 2.3% 1.26 [1.11, 1.43]
Bernstein 2008 (3) 46 59 30 61 1.6% 1.59 [1.19, 2.12]
Bernstein 2008 (4) 18 31 13 30 0.9% 1.34 [0.81, 2.22]
Bezay 2016 24 41 16 40 1.0% 1.46 [0.93, 2.32]
Bologa 2012 19 21 10 19 1.0% 1.72 [1.10, 2.69]
Bresson 2006 108 151 90 149 2.1% 1.18 [1.00, 1.40]
Brooks 2015 26 40 30 40 1.6% 0.87 [0.65, 1.16]
Brown 2010 24 30 16 32 1.2% 1.60 [1.08, 2.36]
Butler 1969 30 56 20 47 1.1% 1.26 [0.83, 1.90]
Chen 2018 12 20 10 20 0.7% 1.20 [0.68, 2.11]
Chichester 2012 13 20 10 20 0.8% 1.30 [0.75, 2.24]
Collier 1979 20 86 11 86 0.6% 1.82 [0.93, 3.56]
Cox 2009 (5) 9 13 5 13 0.5% 1.80 [0.83, 3.92]
Cox 2009 (6) 6 13 9 13 0.6% 0.67 [0.33, 1.33]
Cummings 2014 29 30 27 30 2.2% 1.07 [0.94, 1.23]
Durbin 2020 6 8 4 8 0.4% 1.50 [0.67, 3.34]
Ehrlich 2008 22 85 30 85 1.0% 0.73 [0.46, 1.16]
Ensoli 2009 8 8 7 7 1.8% 1.00 [0.79, 1.27]
EUCTR2009-015103-58-FI 31 33 25 28 2.1% 1.05 [0.90, 1.23]
EUCTR2010-019775-29-AT 349 368 359 382 2.5% 1.01 [0.97, 1.04]
Fries 2017 21 80 45 80 1.1% 0.47 [0.31, 0.71]
Gantt 2018 23 26 21 25 1.9% 1.05 [0.84, 1.31]
Glenn 2013 14 40 9 40 0.5% 1.56 [0.76, 3.18]
Greenberg 2018 187 322 182 328 2.2% 1.05 [0.92, 1.20]
Harro 2012 21 32 17 32 1.1% 1.24 [0.82, 1.86]
Hung 2019 24 46 26 44 1.2% 0.88 [0.61, 1.28]
Jackson 2009 123 164 127 151 2.3% 0.89 [0.80, 1.00]
Juergens 2014 163 271 160 278 2.2% 1.05 [0.91, 1.20]
Kashala 2002 7 8 3 6 0.4% 1.75 [0.75, 4.06]
Kotloff 2001 7 15 6 15 0.4% 1.17 [0.51, 2.66]
Kunz 1976 (7) 44 65 27 65 1.4% 1.63 [1.17, 2.28]
Kunz 1976 (8) 33 48 30 48 1.6% 1.10 [0.82, 1.47]
Landrum 2017 19 22 16 22 1.5% 1.19 [0.88, 1.61]
Langley 2009 57 114 47 112 1.6% 1.19 [0.90, 1.58]
Leroux-Roels 2016b 156 158 152 159 2.5% 1.03 [0.99, 1.07]
Leroux-Roels 2019 72 72 70 72 2.5% 1.03 [0.98, 1.08]
Liang 2010 542 1814 1709 8184 2.4% 1.43 [1.32, 1.55]
Low 2014 38 43 35 39 2.2% 0.98 [0.85, 1.15]
Manoff 2019 7 9 12 18 0.9% 1.17 [0.72, 1.88]
Mark 1994 95 116 85 105 2.3% 1.01 [0.89, 1.15]
Meulen 2015 31 33 25 28 2.1% 1.05 [0.90, 1.23]
NCT00693615 17 18 22 22 2.2% 0.94 [0.81, 1.09]
NCT01447407 35 41 33 40 2.0% 1.03 [0.85, 1.25]
NCT03284710 28 36 15 24 1.3% 1.24 [0.87, 1.78]
Nicholson 2009 60 101 50 99 1.7% 1.18 [0.91, 1.51]
Nolan 2008 162 200 50 200 1.7% 3.24 [2.53, 4.16]
Pan 2013 6 28 9 30 0.4% 0.71 [0.29, 1.75]
Paoletti 2001 9 15 5 15 0.4% 1.80 [0.79, 4.11]
Pillet 2019 48 75 35 75 1.5% 1.37 [1.02, 1.84]
Pressler 1982 9 77 8 83 0.4% 1.21 [0.49, 2.98]
Riddle 2016 10 12 9 12 1.1% 1.11 [0.74, 1.68]
Ruckwardt 2021 37 45 46 50 2.1% 0.89 [0.76, 1.05]
Sheldon 2016 41 73 50 72 1.7% 0.81 [0.63, 1.04]
Tapia 2021 17 145 23 144 0.7% 0.73 [0.41, 1.31]
Verdijk 2013 10 15 10 15 0.9% 1.00 [0.60, 1.66]
Verdijk 2014 14 60 16 60 0.6% 0.88 [0.47, 1.63]
Westritschnig 2014 32 33 30 32 2.3% 1.03 [0.93, 1.15]
Wressnigg 2013 77 151 78 149 1.9% 0.97 [0.78, 1.21]
Wu 2017 63 96 52 97 1.8% 1.22 [0.97, 1.55]
Zhu 2013 107 240 123 240 2.0% 0.87 [0.72, 1.05]
Subtotal (95% CI) 6528 12872 90.3% 1.13 [1.06, 1.20]
Total events 3490 4327
Heterogeneity: Tau² = 0.03; Chi² = 430.78, df = 64 (P < 0.00001); I² = 85%
Test for overall effect: Z = 3.97 (P < 0.0001)
2.21.2 Participants with diagnosis
Brady 2009 13 57 2 60 0.2% 6.84 [1.61, 28.99]
Brady 2009 64 243 38 239 1.3% 1.66 [1.16, 2.37]
Leroux-Roels 2016a 33 60 37 60 1.5% 0.89 [0.66, 1.21]
Longo 2009 10 11 9 11 1.4% 1.11 [0.79, 1.55]
Moustafa 2012 37 74 26 73 1.2% 1.40 [0.96, 2.06]
Rello 2017 95 104 83 98 2.3% 1.08 [0.97, 1.20]
Vandenberghe 2016 18 21 15 16 1.9% 0.91 [0.74, 1.13]
Subtotal (95% CI) 570 557 9.7% 1.17 [0.93, 1.47]
Total events 270 210
Heterogeneity: Tau² = 0.06; Chi² = 26.67, df = 6 (P = 0.0002); I² = 78%
Test for overall effect: Z = 1.35 (P = 0.18)
Total (95% CI) 7098 13429 100.0% 1.13 [1.07, 1.20]
Total events 3760 4537
Heterogeneity: Tau² = 0.03; Chi² = 459.94, df = 71 (P < 0.00001); I² = 85%
Test for overall effect: Z = 4.22 (P < 0.0001)
Test for subgroup differences: Chi² = 0.08, df = 1 (P = 0.78), I² = 0%
0.01 0.1 1 10 100
Aluminium placebo/no intervention
Footnotes
(1) a
(2) b
(3) b
(4) a
(5) a
(6) b
(7) a
(8) b

Supplementary Figure S36. Subgroup analyses of adverse events considered non-serious according to health status of the participants at entry into the trial. Random-effects model.



Supplementary Figure S37. Subgroup analyses of adverse events considered non-serious according to type of pathogen the vaccine is directed against. Random-effects model.

Supplementary Analysis S2.

Individual adverse events considered non-serious analyses

Meta-analysis showed evidence of a harmful effect of aluminium adjuvants compared with placebo or no intervention when assessing the following adverse events considered non-serious:

swelling (RR 1.38, 95% CI 1.17 to 1.62);

injection site pain (RR 1.38, 95% CI 1.19 to 1.61);

tenderness (RR 1.67, 95% CI 1.28 to 2.17);

muscle pain (RR 1.30, 95% CI 1.05 to 1.61);

erythema (RR 1.23, 95% CI 1.02 to 1.50);

musculoskeletal and connective tissue disorders (RR 1.90, 95% CI 1.16 to 3.14);

pain (RR 1.32, 95% CI 1.19 to 1.46);

injection site reaction (RR 2.02, 95% CI 1.23 to 3.32);

myalgia proximal (RR 2.12, 95% CI 1.66 to 2.72); and

rash (RR 1.92, 95% CI 0.99 to 3.72).

Meta-analysis showed no evidence of a difference between aluminium adjuvants versus placebo or no intervention when assessing the following non-serious adverse events: induration, redness, fever onset within 7 days, fever, malaise, fatigue, headache, nausea, vomiting, joint pain, lymph node swelling, itching, sore throat, feeling hot, feverishness, myalgia, ecchymosis, shivering, loss of appetite, arthralgia, stayed home, upper respiratory tract infection, back pain, oropharyngeal pain, hypertension, bruising, cough, hypotension, anorexia, numbness, muscle stiffness, sweating, exanthema, rhinorrhoea, dizziness, nasopharyngitis, urinary tract infection, change in feeding pattern, more crying, infection, bronchitis, injury, poisoning and procedural complications, flu-like symptoms, joint sprain, warmth, insomnia, constipation, toothache, alanine aminotransferase increased, arthropod bite, aspartate aminotransferase increased, contusion, dysphonia, flushing, vertigo positional, vulvovaginal candidiasis, pruritus, pharyngitis, influenza-like illness, rhinitis, chills, gastrointestinal symptoms, blood and lymphatic disorders, gastrointestinal disorders, flatulence, injection site haemorrhage, acute sinusitis, conjunctivitis, herpes zoster, ligament sprain, blood creatinine increased, blood pressure increased, hemoglobin decreased, bursitis, neck pain, tendonitis, nervous system disorders, sinus congestion, skin and subcutaneous disorders, ear and labyrinth disorders, eye disorders, dental caries, food poisoning, gastritis, gingival disorder, edema peripheral, hepato-biliary disorders, renal and urinary disorders, blood bilirubin increased, blood urea increased, epistaxis/nose bleeding, nasal congestion, hardening, less appetite, local administration site disorders, cardio-vascular disorders, general disorders, kidney and urinary disorders, psychiatric disorders, asthma, dyspepsia, cystitis, influenza, oral herpes, hyperglycaemia, ear infection, reproductive system and breast disorders, swelling in the neck, asthenia, diarrhoea, and gastroenteritis.

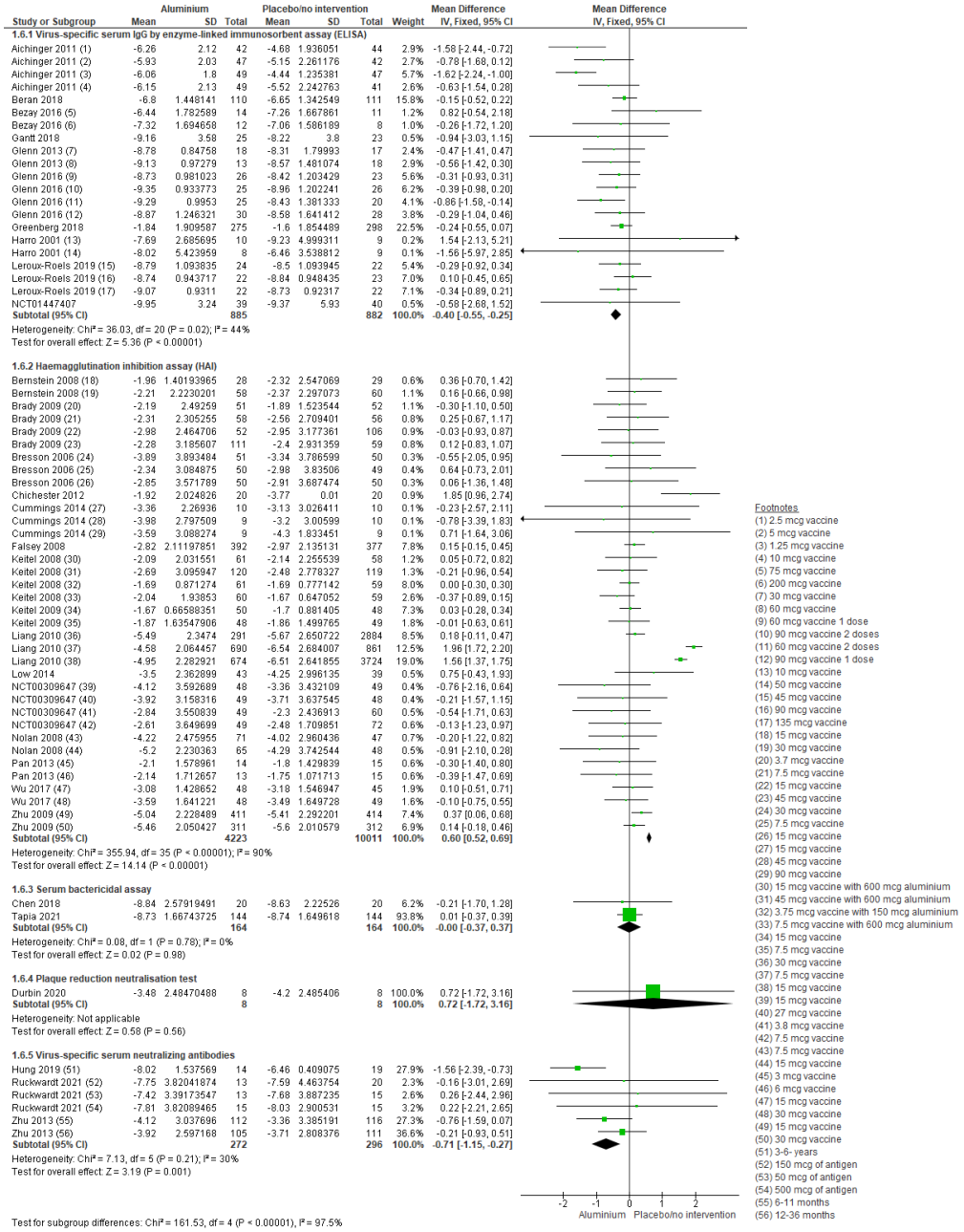
Individual non-serious adverse events reported by only one trial that could not be meta-analysed are shown in the table below.

Non-serious adverse events reported in the aluminum-adjuvanted groups versus the placebo group or no intervention		
Non-serious adverse events	**Aluminium event/total (study ID)**	**Placebo event/total (study ID)**
Sinus headache	1/32 (Harro 2012)	0/32 (Harro 2012)
Body pain	2/60 (Basavaraj 2014)	1/60 (Basavaraj 2014)
Common cold	2/60 (Basavaraj 2014)	1/60 (Basavaraj 2014)
Allergy on face	1/60 (Basavaraj 2014)	0/60 (Basavaraj 2014)
Burning sensation	1/60 (Basavaraj 2014)	0/60 (Basavaraj 2014)
Lower limb pain	0/60 1/20 (Basavaraj 2014)	1/60 0/20 (Basavaraj 2014)
Systolic hypertension	1/80 (Fries 2017)	4/80 (Fries 2017)
Abnormal crying	9/40 (Brooks 2015)	9/40 (Brooks 2015)
Respiratory symptoms	5/14 0/13 (Cox 2009)	0/14 3/13 (Cox 2009)
Contralateral inflammation	2/8 (Kashala 2002)	1/6 (Kashala 2002)
Joint swelling	1/15 (Kotloff 2001)	0/15 (Kotloff 2001)
Limitation of movement	13/43 (Low 2014)	13/39 (Low 2014)
Nodule	10/33 (Meulen 2015)	3/28 (Meulen 2015)
Muscle rigidity	1/60 (Verdrijk 2014)	1/60 (Verdrijk 2014)
Change in stool	12/60 (Verdrijk 2014)	9/60 (Verdrijk 2014)
Change in sleeping pattern	14/60 (Verdrijk 2014)	16/60 (Verdrijk 2014)
Change in behavior	10/60 (Verdrijk 2014)	11/60 (Verdrijk 2014)
Pain in extremity	7/33 (Westritschnig 2014)	0/32 (Westritschnig 2014)
Eclampsia	1/120 (Zhu 2013)	2/120 (Zhu 2013)
Exanthema	2/46 (Hung 2019)	1/44 (Hung 2019)
Blood fibrinogen decreased	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Blood lactate dehydrogenase increased	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Bone contusion	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Excoriation	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Infected dermal cyst	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Platelet count increased	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Rosacea	1/60 (Leroux-Roels 2016b)	0/60 (Leroux-Roels 2016b)
Injury	1/18 (NCT00693615)	6/22 (NCT00693615)
Infection viral	1/20 (Chen 2018)	0/20 (Chen 2018)
Stress	0/60 (Leroux-Roels 2016b)	1/60 (Leroux-Roels 2016b)
Myalgia - distal	39 /200 (Nolan 2008)	35/200 (Nolan 2008)
Cardiac disorders	1/73 (Sheldon 2016)	2/72 (Sheldon 2016)
Palpitations	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Abdominal distension	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Infected bites	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Fall	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Periorbital hematoma	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Post-traumatic neck syndrome	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Aspartate aminotransferase increased	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Blood cholesterol increased	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Dyslipidemia	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Musculoskeletal and connective tissue disorders	5/73 (Sheldon 2016)	5/72 (Sheldon 2016)
Neoplasms benign malignant and unspecified (including cysts and polyps)	1/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Limb discomfort	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Heart valve stenosis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Vertigo postural	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Endocrine disorders	2/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Hypothyroidism	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Testicular failure	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Vitreous detachment	0/73 (Sheldon 2016)	2/72 (Sheldon 2016)
Cellulitis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Cholelithiasis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Folliculitis	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Hand fracture	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Osteoporosis	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Urethral stenosis	1/73 (Sheldon 2016)	0/72 (Sheldon 2016)
Prostatism	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Blood cortisol increased	0/73 (Sheldon 2016)	1/72 (Sheldon 2016)
Local reactions	118/770 (Zhu 2009 all)	55/770 (Zhu 2009 all)
Neurologic	5/770 (Zhu 2009 all)	4/770 (Zhu 2009 all)
Systemic	65/770 (Zhu 2009 all)	60/770 (Zhu 2009 all)
Bacterial vaginosis	1/20 (Chen 2018)	0/20 (Chen 2018)
Blood pressure decreased	1/30 (Atsmon 2014)	0/30 (Atsmon 2014)
Herpangina	1/46 (Hung 2019)	0/44 (Hung 2019)
Papules	0/46 (Hung 2019)	1/44 (Hung 2019)
Lymph node pain	1/32 (Harro 2012)	0/32 (Harro 2012)

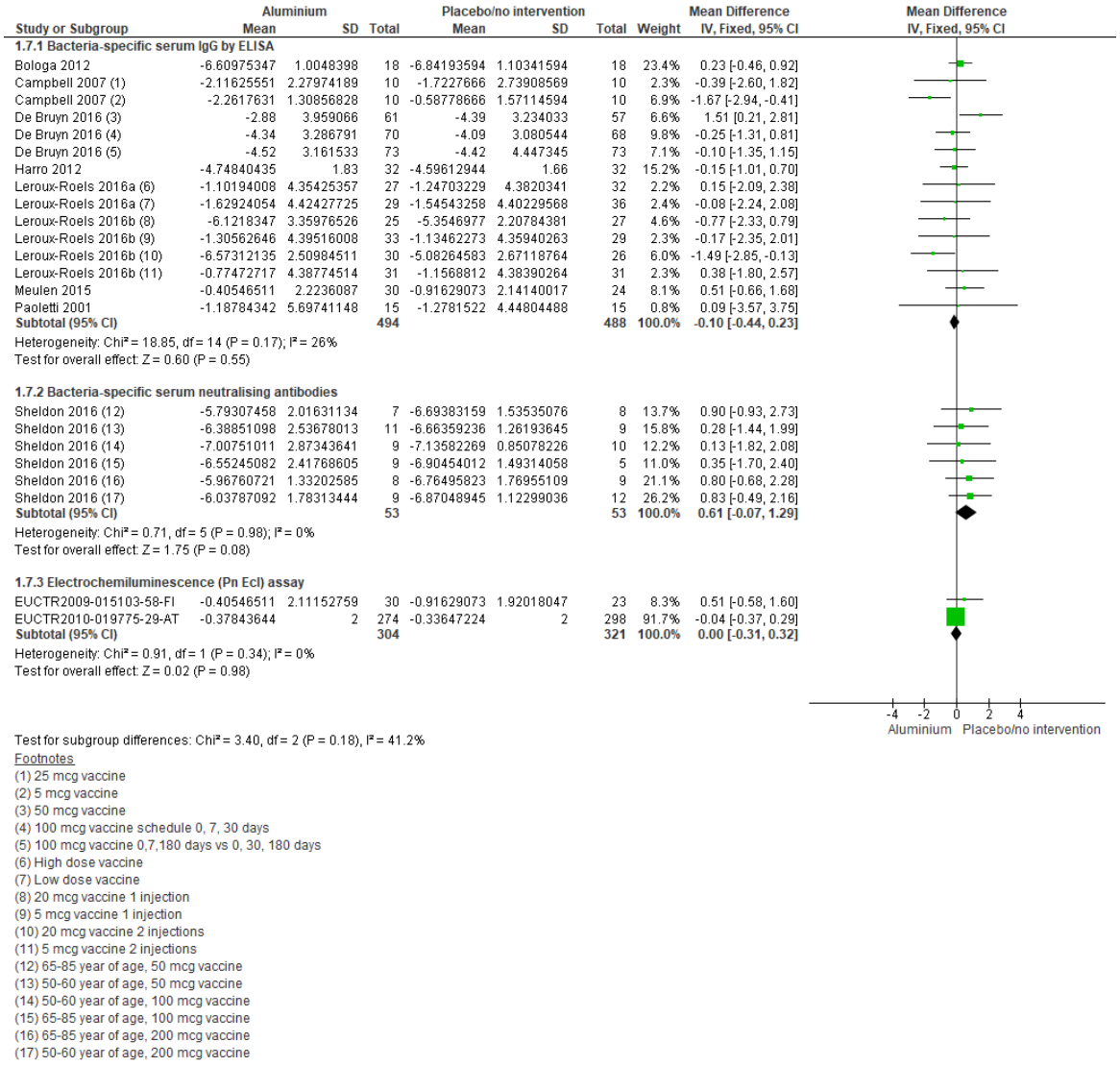
continues below

Tooth disorder	2/72 (Leroux-Roels 2019)	0/72 (Leroux-Roels 2019)
Tinea Pedis	2/72 (Leroux-Roels 2019)	0/72 (Leroux-Roels 2019)
Wheezing	5/72 (Leroux-Roels 2019)	5/72 (Leroux-Roels 2019)
Hemoglobin abnormal	2/22 (Landrum 2017)	1/22 (Landrum 2017)
Blood LDH increase	1/22 (Landrum 2017)	1/22 (Landrum 2017)
Whole blood count increased	2/22 (Landrum 2017)	3/22 (Landrum 2017)
Neutrophil count increased	1/22 (Landrum 2017)	2/22 (Landrum 2017)
Hypoglycaemia	0/22 (Landrum 2017)	1/22 (Landrum 2017)
Blood sodium increased	2/22 (Landrum 2017)	1/22 (Landrum 2017)
Teething	0/33 (Meulen 2015)	1/28 (Meulen 2015)
Laryngitis	0/33 (Meulen 2015)	1/28 (Meulen 2015)
Respiratory syncytial virus bronchiolitis	1/33 (Meulen 2015)	0/28 (Meulen 2015)
Dysmenorrhea	0/18 (NCT00693615)	2/22 (NCT00693615)
Flu syndrome	1/18 (NCT00693615)	2/22 (NCT00693615)
Depression	0/18 (NCT00693615)	1/22 (NCT00693615)
Urogenital neoplasia	0/18 (NCT00693615)	1/22 (NCT00693615)
Vaginal moniliasis	0/18 (NCT00693615)	1/22 (NCT00693615)
Injection site vesicles	0/8 (Durbin 2020)	1/8 (Durbin 2020)
Armpit swelling	0/25 (Gantt 2018)	1/23 (Gantt 2018)
Gastroesophageal reflux	1/25 (Gantt 2018)	0/23 (Gantt 2018)
Herpes simplex	0/25 (Gantt 2018)	2/23 (Gantt 2018)
Laceration	0/25 (Gantt 2018)	1/23 (Gantt 2018)
Otitis	3/33 (Meulen 2015)	1/28 (Meulen 2015)
Head injury	1/33 (Meulen 2015)	0/28 (Meulen 2015)
Bacterial vaginosis	0/36 (NCT03284710)	2/24 (NCT03284710)
Orchitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Muscle twitching	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Dysuria	1/36 (NCT03284710)	0/24 (NCT03284710)
Otitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Discomfort	6/119 (Mark 1994)	4/105 (Mark 1994)
Neutropenia	3/45 (Ruckwardt 2021)	3/50 (Ruckwardt 2021)
Eosinophilia	1/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Leukopenia	1/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Lymphopenia	0/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Bradycardia	2/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Tachycardia	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Burns first degree	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Hepatic enzyme increased	1/45 (Ruckwardt 2021)	0/50 (Ruckwardt 2021)
Abnormal dreams	0/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)
Laryngeal pain	0/45 (Ruckwardt 2021)	2/50 (Ruckwardt 2021)
Body tinea	0/36 (NCT03284710)	1/24 (NCT03284710)
Fungal skin infection	0/36 (NCT03284710)	1/24 (NCT03284710)
Furuncle	0/36 (NCT03284710)	1/24 (NCT03284710)
Genitourinary chlamydia infection	5/36 (NCT03284710)	0/24 (NCT03284710)
Genitourinary tract gonococcal infection	1/36 (NCT03284710)	1/24 (NCT03284710)
Pelvic inflammatory disease	1/36 (NCT03284710)	0/24 (NCT03284710)
Subcutaneous abscess	1/36 (NCT03284710)	0/24 (NCT03284710)
Tooth infection	0/36 (NCT03284710)	1/24 (NCT03284710)
Vulvovaginitis trichomonal	1/36 (NCT03284710)	0/24 (NCT03284710)
Arthropod sting	1/36 (NCT03284710)	0/24 (NCT03284710)
Limb injury	0/36 (NCT03284710)	1/24 (NCT03284710)
Skin abrasion	0/36 (NCT03284710)	1/24 (NCT03284710)
Sunburn	1/36 (NCT03284710)	0/24 (NCT03284710)
neutrophil count decreased	1/36 (NCT03284710)	0/24 (NCT03284710)
Abnormal loss of weight	0/36 (NCT03284710)	1/24 (NCT03284710)
Musculoskeletal pain	0/36 (NCT03284710)	1/24 (NCT03284710)
Glycosuria	1/36 (NCT03284710)	1/24 (NCT03284710)
Dysfunctional uterine bleeding	1/36 (NCT03284710)	0/24 (NCT03284710)
Prostatitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Vaginal discharge	1/36 (NCT03284710)	0/24 (NCT03284710)
Hand dermatitis	1/36 (NCT03284710)	0/24 (NCT03284710)
Rash maculo-papular	1/36 (NCT03284710)	0/24 (NCT03284710)
Rash papular	1/36 (NCT03284710)	0/24 (NCT03284710)
Rash pruritic	1/36 (NCT03284710)	0/24 (NCT03284710)
Infection	4/36 (NCT03284710)	11/24 (NCT03284710)
Impetigo	1/145 (Tapia 2021)	1/144 (Tapia 2021)
Pharyngitis	5/145 (Tapia 2021)	4/144 (Tapia 2021)
Varicella	0/145 (Tapia 2021)	1/144 (Tapia 2021)
Thermal burn	1/145 (Tapia 2021)	0/144 (Tapia 2021)
erythema/induration	9/36 (NCT03284710)	6/24 (NCT03284710)
malaise and or fatigue	8/36 (NCT03284710)	3/24 (NCT03284710)
Max. Systemic Symptoms	13/36 (NCT03284710)	7/24 (NCT03284710)
Ear pain	0/36 (NCT03284710)	0/24 (NCT03284710)
Abdominal pain lower	0/36 (NCT03284710)	0/24 (NCT03284710)
Peripheral swelling	0/36 (NCT03284710)	0/24 (NCT03284710)
Vessel puncture site phlebitis	0/36 (NCT03284710)	0/24 (NCT03284710)
Lower respiratory tract infection	0/36 (NCT03284710)	0/24 (NCT03284710)
Respiratory tract infection	0/36 (NCT03284710)	0/24 (NCT03284710)
Urinary tract infection bacterial	0/36 (NCT03284710)	0/24 (NCT03284710)
Vulvovaginitis gonococcal	0/36 (NCT03284710)	0/24 (NCT03284710)
Eye injury	0/36 (NCT03284710)	0/24 (NCT03284710)
Human bite	0/36 (NCT03284710)	0/24 (NCT03284710)
Skin laceration	0/36 (NCT03284710)	0/24 (NCT03284710)
Investigation	14/36 (NCT03284710)	6/24 (NCT03284710)
Cardiac murmur	14/36 (NCT03284710)	6/24 (NCT03284710)
Blood bicarbonate decreased	14/36 (NCT03284710)	6/24 (NCT03284710)
Platelet count decreased	14/36 (NCT03284710)	6/24 (NCT03284710)
White blood cell count decreased	14/36 (NCT03284710)	6/24 (NCT03284710)
Vulvovaginal rash	14/36 (NCT03284710)	6/24 (NCT03284710)
Rhinalgia	14/36 (NCT03284710)	6/24 (NCT03284710)
Dermatitis	14/36 (NCT03284710)	6/24 (NCT03284710)
Ingrowing nail	14/36 (NCT03284710)	6/24 (NCT03284710)
Musculoskeletal chest pain	0/45 (Ruckwardt 2021)	1/50 (Ruckwardt 2021)

Supplementary Table S3. List of individual adverse events considered non-serious reported in only one trial.



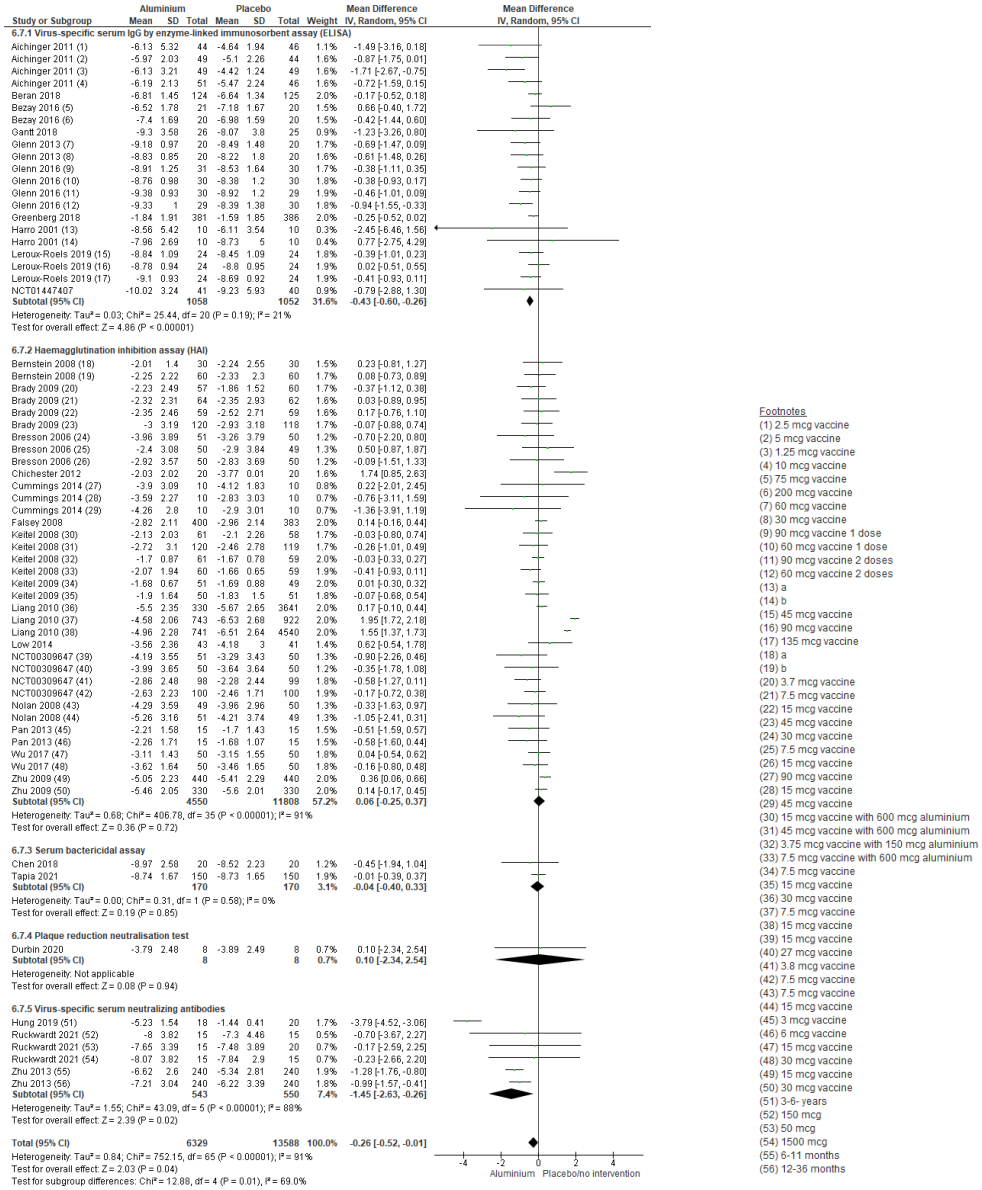
Supplementary Figure S38. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean titres (GMT) grouped by analytical assay. Fixed-effect model.



Supplementary Figure S39. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the geometric mean concentrations (GMC) grouped by analytical assay. Fixed-effect model.

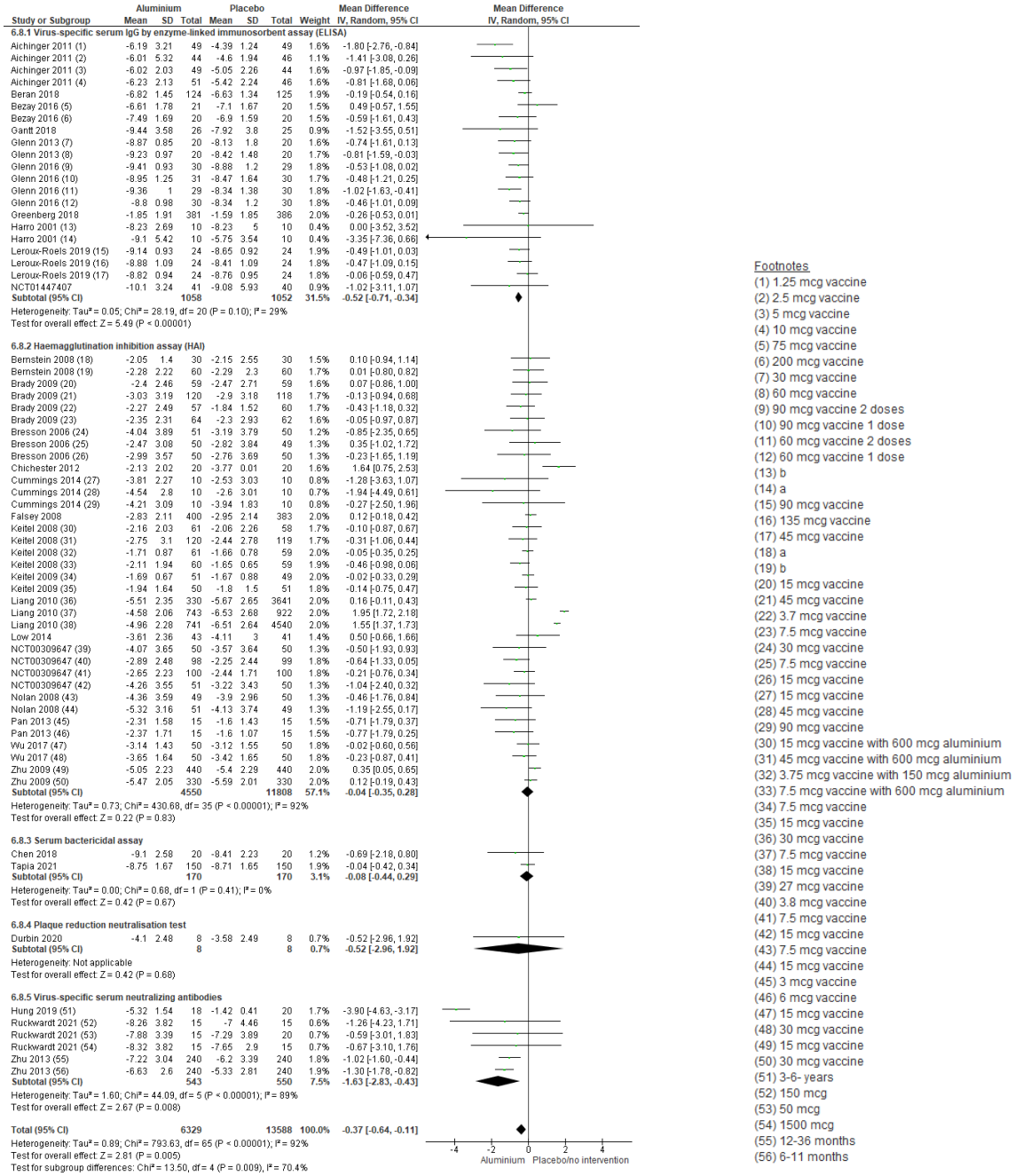
Supplementary Figure S40. Overview of the serology analysis with subgroup analysis. We performed subgroup analyses on the most frequently reported analytical assays: enzyme-linked immunosorbent assay (ELISA) and haemagglutination inhibition assay (HAI). We observed a beneficial effect of aluminium adjuvants on serological response in sporadic subgroups (based on vaccine types and on pathogen types).

		SEROLOGY							
		Geometric Mean Titres (GMT) analysis per assay					Geometric Mean Concentration (GMC) analysis per assay		
		ELISA n=10 Favors aluminium (MD -0.47, 95% CI -0.69 to -0.24; participants = 1767; I2 = 44%)	HAI n=15 No effect	Serum bactericidal assay n=1 No effect	Plaque reduction neutralisation test n=1 No effect	Virus-specific serum neutralizing antibodies n=3 Favors aluminium (MD -0.69, 95% CI -1.27 to -0.10; participants = 568; I2 = 30%)	ELISA n=10 No effect	Bacteria-specific serum neutralizing antibodies n=1 No effect	Electrochemiluminescence (Pn EcI) assay n=2 No effect
Subgroup analysis									
Risk of bias	Low High	No effect	No effect	n.a.	n.a.	n.a.	No effect	n.a.	n.a.
Vested interest	Low High	No effect	No effect	n.a	n.a	n.a	No effect	n.a	n.a
Aluminium type	Adju-Phos® Alhydrogel Aluminium phosphat Aluminium potassiur Aluminiuminium hyd Not reported	No effect	No effect	n.a	n.a	n.a	No effect	n.a	n.a
Vaccine type **Test for subgroup differences: P = 0.03; I^2 = 57.4%**	Influenza *C. difficile* Cytomegalovirus Human papillomaviru		No effect	n.a	n.a	n.a	No effect	n.a	n.a
	Respiratory syncytial	Favors aluminium n=4							
	Ross River Virus	Favors aluminium n=1							
	S.Aureus and Candida *Streptococcus pneumoniae*								
Age	Children Adults Elderly Mixed	No effect	No effect	n.a	n.a	n.a	No effect	n.a	n.a
Follow up	Short Medium	No effect	No effect	n.a.	n.a.	n.a.	No effect	n.a.	n.a.
Health	Healthy With diagnosis	No effect	No effect	n.a.	n.a.	n.a.	No effect	n.a.	n.a.
Pathogen type	Intracellular Extracellular	No effect	n.a.	n.a.	n.a.	n.a.	No effect	n.a.	n.a.

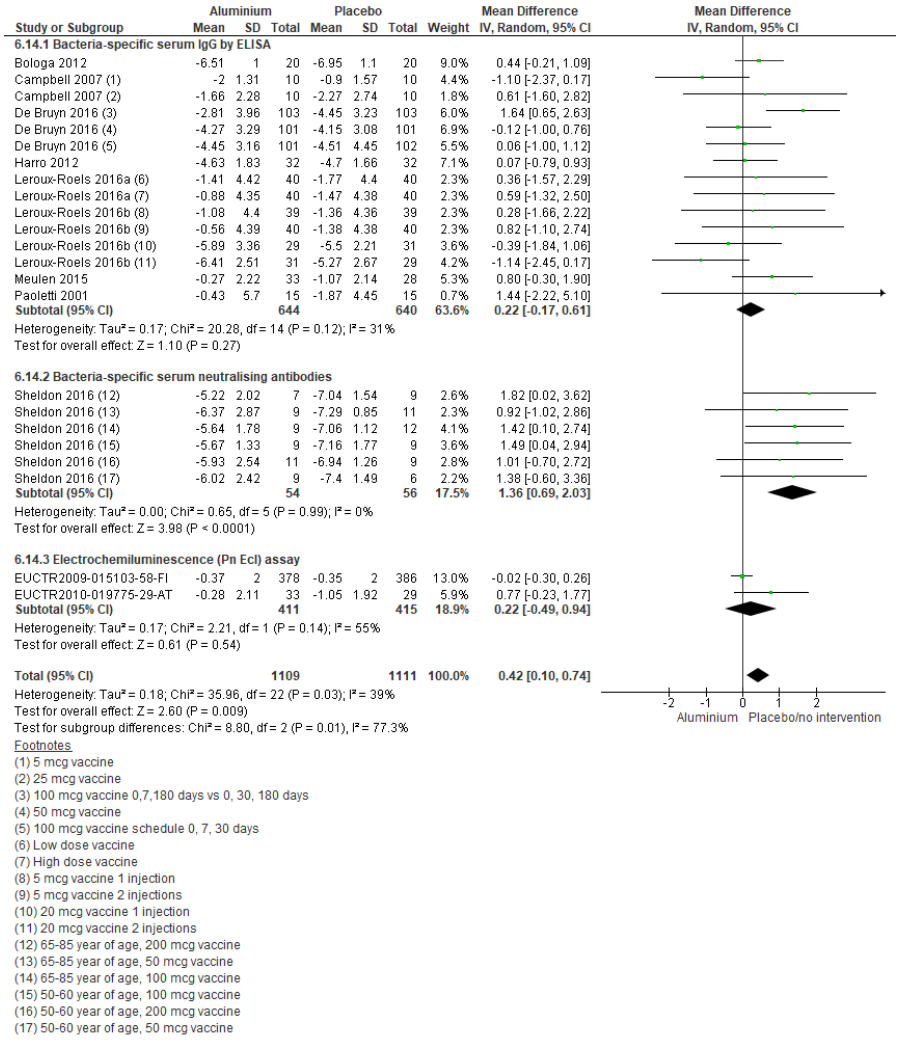


Supplementary Figure S41. Sensitivity analysis. 'Best worst' case scenario with 1 standard deviation for serology analysed with geometric mean titre (GMT). 777/3950 (19.7%) participants in the intervention group versus 2231/10,217 (21.8%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'best worst' case scenario, so that there was a beneficial effect of aluminium adjuvants in random-effects meta-analyses ('Best worst' with 1 standard deviation: MD -0.26, 95% CI -0.52 to -0.01; participants = 19 917; studies = 66; I^2 = 91%. 'Best worst' with 2 standard deviations: MD -0.37, 95% CI -0.64 to -0.11; participants = 19 917; studies = 66; I^2 = 92%). Random-effects model.

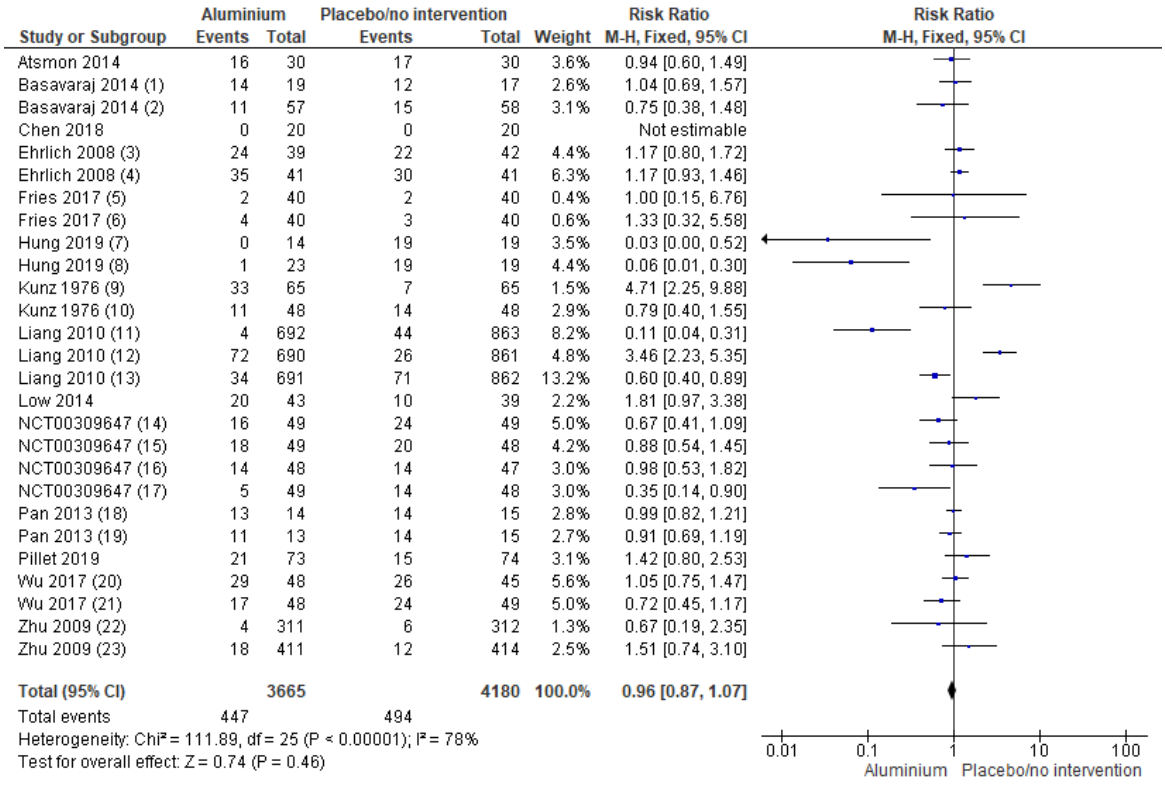
Incomplete outcome data did not have the potential to influence the result in the 'worst best' case scenario analyses with 1 or 2 standard deviations (data not shown).



Supplementary Figure S42. Sensitivity analysis. 'Best worst' case scenario with 2 standard deviations for serology analysed with geometric mean titre (GMT).



Supplementary Figure S43. Sensitivity analysis. 'Worst best' case scenario with 2 standard deviations for serology analysed with geometric mean concentration (GMC). 258/784 (32.9%) participants in the intervention group versus 249/478 (52.1%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the 'worst best' case scenario, so that there was a harmful effect of aluminium adjuvants in random-effects meta-analysis (2 standard deviations: MD 0.42, 95% CI 0.10 to 0.74; participants = 2220; studies = 23; I^2 = 39%). Incomplete outcome data did not have the potential to influence the result in the 'worst-best' case analysis with 1 standard deviation. Incomplete outcome data did not have the potential to influence the result in the 'best worst' case scenario analysis. Meta-analyses of trials with imputed SD showed no evidence of a difference between aluminium adjuvants versus placebo (GMT: MD -0.11, 95% CI -0.33 to 0.12; GMC: MD 0.04, 95% CI -0.17 to 0.26).



Footnotes
(1) vaccine with or without aluminium
(2) placebo with or without aluminium
(3) 7.5 mcg vaccine
(4) 15 mcg vaccine
(5) 60 mcg vaccine
(6) 90 mcg vaccine
(7) 3-6 years of age
(8) 2-35 months of age
(9) subunit vaccine
(10) whole virus vaccine
(11) 30 mcg vaccine
(12) 7.5 mcg vaccine
(13) 15 mcg vaccine
(14) 15 mcg vaccine
(15) 27 mcg vaccine
(16) 3.8 mcg vaccine
(17) 7.5 mcg vaccine
(18) 3 mcg vaccine
(19) 6 mcg vaccine
(20) 15 mcg vaccine
(21) 30 mcg vaccine
(22) 30 mcg vaccine
(23) 15 mcg vaccine

Supplementary Figure S44. Meta-analysis of the effect of aluminium adjuvants compared with placebo or no intervention on the outcome seroprotection. Fixed-effect model.

Supplementary Analysis S3.

Subgroup and sensitivity analyses for the outcome seroprotection

Test for subgroup differences showed no difference when comparing trials at high risk of bias to trials at low risk of bias; trials at risk of vested interest to trials at no risk of vested interest; trials with different aluminium adjuvant types; and trials with vaccines against different pathogen types.

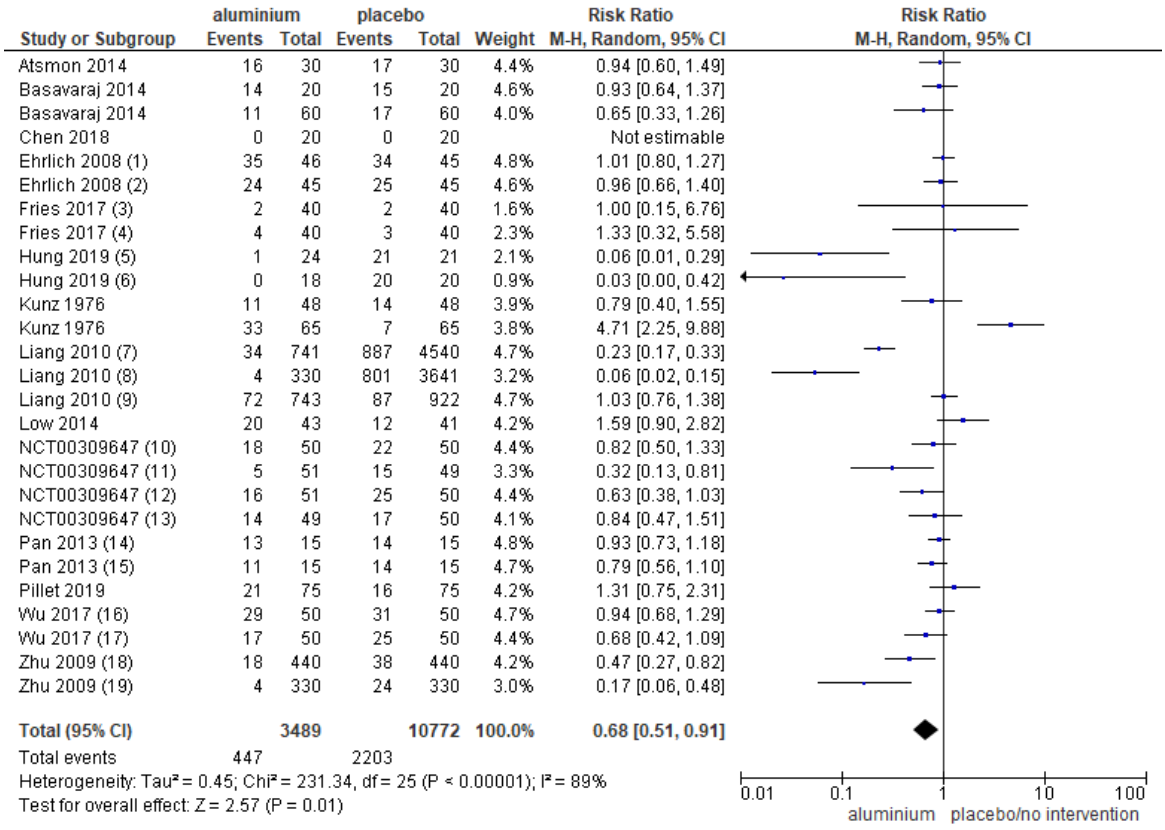
Test for subgroup differences was statistically significant (P = 0.0005) in the subgroup analysis according to vaccine types.

Test for subgroup differences was statistically significant (P = 0.0004) in the subgroup analysis according to age groups.

Trial Sequential Analysis showed that the accrued information for seroprotection was below 5% of the diversity-adjusted required information size (n = 52 586; Pc 11%, RRR 20%, alpha 2.5%, beta 10%, diversity 82.6%). Accordingly, the analysis was not drawn by the program.

242/3165 (7.6%) participants in the aluminium adjuvant group versus 1709/9986 (17.1%) participants in the control group were lost to follow up. Incomplete outcome data alone seemed to have the potential to influence the result in the ‘best worst’ case scenario analysis.

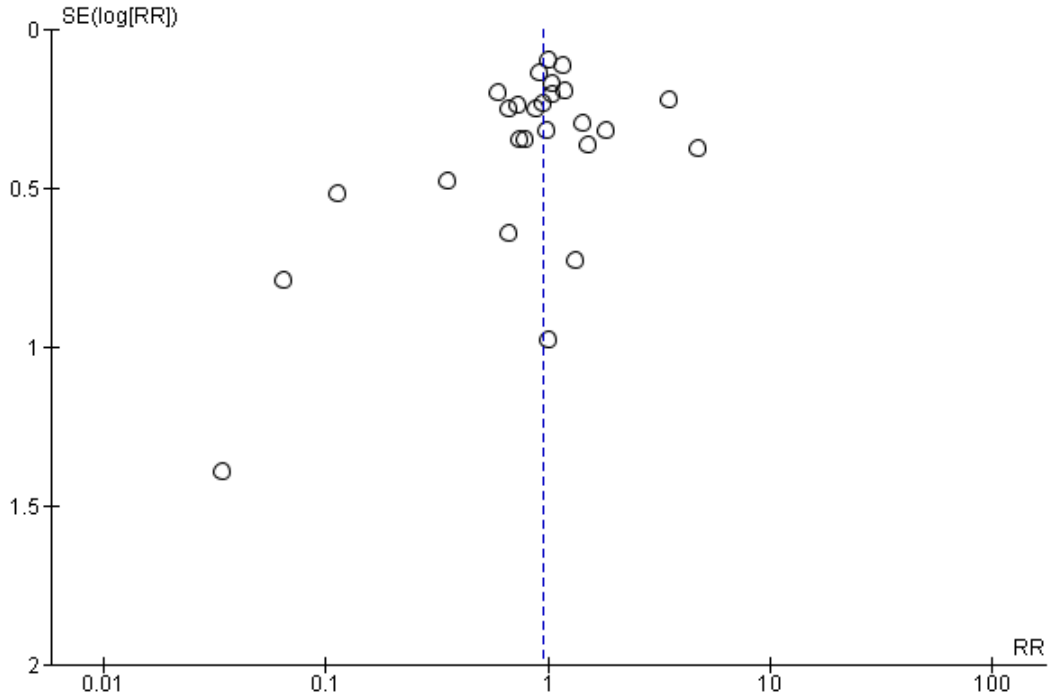
The 'worst best' case scenario analysis showed that incomplete outcome data did not have the potential to influence the result.



Footnotes
(1) 7.5 mcg vaccine
(2) 15 mcg vaccine
(3) 90 mcg vaccine
(4) 60 mcg vaccine
(5) 3-6 years of age
(6) 2-35 months of age
(7) 15 mcg vaccine, adults
(8) 7.5 mcg vaccine, adults
(9) 15 mcg vaccine, elderly
(10) 7.5 mcg vaccine
(11) 27 mcg vaccine
(12) 15 mcg vaccine
(13) 3.8 mcg vaccine
(14) 3 mcg vaccine
(15) 6 mcg vaccine
(16) 15 mcg vaccine
(17) 30 mcg vaccine
(18) 15 mcg vaccine
(19) 30 mcg vaccine

Supplementary Figure S45. Sensitivity analysis. 'Best worst' case scenario for seroprotection.

Incomplete outcome data did not have the potential to influence the result in the 'worst best' case scenario analysis of seroprotection (data not shown).



Supplementary Figure S46. Funnel plot for seroprotection outcome. Visual inspection of funnel plot may suggest potential publication bias for smaller trials reporting a harmful effect of aluminium adjuvants in the intervention group. Regression-based Harbord test showed no small-study effects (beta = 0.99).

Supplementary Analysis S4.

Recency of the different trials

We performed our analysis using Review Manager 5.4. In this software, we used the inbuilt function of sorting the forest plots entries by year and we visually inspected the forest plots to account for the recency of different trials. Together with the inspection of the I^2, we did not observe an influence on the analysis stemming from the date of when the studies were performed.

Supplementary methods

For the serological response, we used the geometric mean concentration or titre (GMC or GMT) extracted from the first assay reported in the journal publication by the trialists. If a trial tested multiple antigens, we included data from the first antigen of the first assay as presented in the publication. For the meta-analysis, we transformed the GMT in the following way: we calculated the natural Log for both the GMT (or GMC) and for the confidence intervals (CI).

If the standard deviations (SD) were not reported, we calculated them using data from the trial, if possible. When we calculated the SD from the CI, we first calculated a t value from the number of participants (n) analysed for that outcome (t value=TINV(1-0.95; n-1), and then we calculated the SD as follows: (SQRT(n))*(LogCIhigh-LogCIlow) / t value). In the meta-analysis, we then plotted the GMT (or GMC) multiplied by -1 and the SD calculated as described above. When imputing the SD due to lack of reporting of CI by the trialists, an average SD was calculated from the other trials and used in the meta-analysis and presented in sensitivity analyses.

We did not impute missing values for any outcomes in our primary analyses. In our sensitivity analysis for dichotomous and continuous outcomes, we did impute data.

Unit of analysis issues

We included data from trials where participants were individually randomised to two or more intervention groups. When trials had more than two intervention groups, we considered the following scenarios:

Experimental intervention	Control
Vaccine with different antigen concentrations and same aluminium adjuvant concentration	Same vaccine with different antigen concentrations without aluminium adjuvant
Vaccine with different antigen concentration and different aluminium adjuvant concentration	Same vaccine with different antigen concentration without aluminium adjuvant
Different aluminium adjuvant concentrations	Placebo

We analysed the scenarios above as follows:

- when the intervention groups had different antigen concentration but same aluminium adjuvant concentration, we pooled the data from all the intervention groups for the safety analyses, and we analysed each intervention group separately for serology.

- when the intervention groups had different antigen concentration and different aluminium adjuvant concentrations, we analysed each intervention group separately both for safety and for serology.

- when we presented analyses from multiple groups from the same trial, the ID of the same trial appears multiple times in the forest plot together with a footnote describing the difference in the vaccine composition (i.e. antigen concentration) in that trial group.

Differences between protocol and review

We did not specifically search for observational studies (quasi-randomised studies; cohort studies; and patient series), and we abandoned our potential plan to provide a narrative account of such data. If and when certainty of benefits of adding aluminium adjuvants to vaccines one day may be identified on patient-relevant outcomes, observational studies should be scrutinized for rare and late occurring harms.

We originally planned to base our results, conclusions and summary of findings (SoF) table on trials at low risk of bias. However, due to lack of relevant data, we based our results, conclusions and SoF table on all trials regardless of the bias assessment.

We originally planned to perform the subgroup analysis on the effects of nanoparticle aluminium adjuvant compared with microparticle aluminium adjuvant regardless of type of aluminium salt. However, due to lack of data, we could not perform such analyses.

We originally planned to perform the subgroup analysis on the effect of total administration of aluminium between trials. However, this was replaced by a meta-regression analysis.

We originally planned to perform the subgroup analysis on the effects of aluminium adjuvant administration between trials with different maximal follow-up periods, by defining the follow up period 'from randomisation until the end of follow up'. However, we decided to change the definition of the follow up period to 'from end of the vaccination until the end of follow up'.

We added a subgroup analysis for trials assessing vaccines against different pathogen types in response to a peer reviewer. We did not describe this analysis in our protocol.

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Manuscript page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Manuscript page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Manuscript page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Manuscript page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Manuscript page 4; Supplementary page 50; Protocol
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Manuscript page 5; Supplementary Table 1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary Table 1;
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Manuscript page 5,6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Manuscript page 5,6; Protocol
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Manuscript page 5,6; Protocol; Supplementary Page 51
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Protocol; Supplementary Page 50
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Manuscript page 6; Protocol
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Manuscript page 6; Supplementary Page 50; Protocol
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Manuscript page 4; Supplementary page 50; Protocol
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing	Supplementary page 50

Section and Topic	Item #	Checklist item	Location where item is reported
		summary statistics, or data conversions.	
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Supplementary Table2,3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Manuscript page 4,5; Supplementary page 50; Protocol
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Manuscript page 5,6; 7 Protocol
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Manuscript page 4,5; Supplementary page 50; Protocol
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Manuscript page 5; Supplementary page 50; Protocol
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Manuscript page 7; Protocol
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Manuscript Page 7; Supplementary Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Appendix 1
Study characteristics	17	Cite each included study and present its characteristics.	Appendix 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary Figure 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Manuscript Page 11 to 17
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	n.a.
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Manuscript Page 11 to 17 Supplementary material for subgroup analysis
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Manuscript Page 11 to 17; Supplementary material for subgroup analysis
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Manuscript Page 11 to 17; Supplementary material for sensitivity analysis
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Manuscript Page 11 to 17; Supplementary material

Section and Topic	Item #	Checklist item	Location where item is reported
			for sensitivity analysis
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Table 1
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Manuscript page 18 to 22
	23b	Discuss any limitations of the evidence included in the review.	Manuscript page 18 to 22
	23c	Discuss any limitations of the review processes used.	Manuscript page 18 to 22
	23d	Discuss implications of the results for practice, policy, and future research.	Manuscript page 18 to 22
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	abstract
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	abstract
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Supplementary page 51
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Manuscript page 23,24
Competing interests	26	Declare any competing interests of review authors.	Manuscript page 25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	n.a.

Supplementary Table S4. PRISMA checklist.